THE SPECTRE KNOT: ONE ENDLESS STRAND ON AN APERIODIC MONOTILE

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Abstract. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss tiles the plane, but only non-periodically. We decorate its straight-edged form Tile(1, 1) with fourteen ribbon pieces, the same on every tile, so that ribbons continue straight across every tile edge. We prove that the pieces never close up: in every tiling by copies of Tile(1, 1) of one handedness whose supertile hierarchy covers the plane, the ribbons contain no closed strand. In the tiling grown from a nested chain of Δ-supertiles, all the pieces of all the tiles join into a single strand, a two-way infinite path that passes through every tile. We do not claim this for other tilings, and computations indicate that it fails in some. The proof shows that every supertile, at every level, connects the ribbon ends on its boundary in the same simple pattern: a hook at every boundary vertex but two, and one crossover. Where supertiles are glued, the ribbons zigzag along the shared boundary, so lengthening it by two edges changes nothing; and the side lengths at every level are reached from those at level 1 in steps of two, because every side length is a Fibonacci number, twice one, or one plus or minus two, with the same parity at every level. This reduces the infinite problem to finitely many cases at level 1, which we check by computer. The single strand is reduced in the same way to a map, computed at level 1, that says where the ribbons of a supertile go one level up. The combinatorial core of the proof, with these computations, is also formalised in the Lean proof assistant. The proof, the programs, the formalisation and this text were produced by the author working with an AI system (see *Use of AI and verification status* in Section 1).

1. Introduction

A *Celtic knot* is a pattern of interlaced ribbons [2, 5]. How many separate strands a finite design of this kind contains has been studied, for instance, by Fisher and Mellor [8]. A *knot tile* is a polygon carrying ribbon segments that end on its edges, so that copies of the tile, placed edge to edge, link up into longer ribbons. We ask for a knot tile whose ribbons, in a tiling of the whole plane, form one endless strand. In a periodic tiling this cannot happen: if the decorated tiling is invariant under a non-zero translation, its ribbons never form a single strand (Remark 3.5). So a tile that only tiles non-periodically is the natural place to look.

In 2023 Smith, Myers, Kaplan and Goodman-Strauss described the *hat*, an aperiodic monotile [18], and then the *Spectre* [19]: a shape that tiles the plane, admits only non-periodic tilings, and admits only tilings in which all tiles have the same handedness, even when reflections are allowed [19, Thm. 2.2]. For background on tilings and substitution see [11, 1]. The Spectre is a curved-edge form of the polygon Tile(1, 1) (Fig. 1): tilings by Spectres correspond exactly to the tilings by Tile(1, 1) in which all tiles have the same handedness [19, §2].

This paper describes a knot tile on the Spectre and proves that it is *mono-stranded* in the following sense: in one explicit tiling, all its pieces form a single strand (Theorem B), and no tiling whose hierarchy covers the plane contains a closed strand (Theorem A). We do not claim one strand for every tiling; computations indicate that some tilings have three (Remark 10.1). The decoration (Fig. 1) puts one crossing point on each of the fourteen edges and joins the 28 ribbon ends inside the tile by fourteen pieces: twelve *hooks*, each turning

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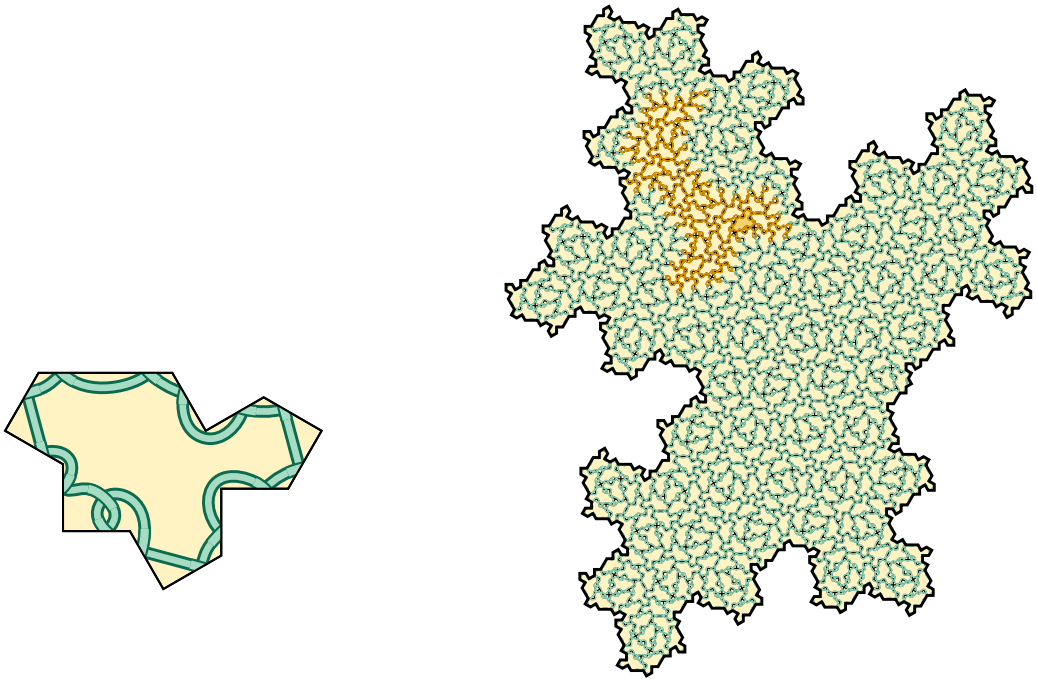


FIGURE 1. Left: the tile Tile(1, 1) with its fourteen ribbon pieces. Right: a supertile of 559 tiles (the block N_3 of Section 8). The original tile N_0 is shaded; all fourteen of its pieces already lie on a single arc of this block, drawn in orange.

back round one vertex, and two pieces that cross each other inside the tile. Across an edge, the ribbons simply continue straight through the crossing point.

Our first result concerns all tilings with a well-behaved hierarchy. Every tiling by Tile(1, 1) with tiles of one handedness (a *homochiral* tiling; Section 2) groups uniquely into a hierarchy of larger and larger *supertiles* [19, Thm. 2.2]. We say that the hierarchy *covers the plane* if the supertiles containing some (equivalently, every) tile exhaust the plane; equivalence holds because, once a supertile containing one tile contains another, the two tiles have the same supertiles from that level on. If it does not, the tiling splits into two or more infinite supertiles (unions of nested chains) with disjoint interiors, separated by a *fault*. (Each tile lies in exactly one supertile of each level, so two such unions are equal or disjoint.)

Theorem A (The Daedalus Theorem)**.** *In every homochiral tiling of the plane by* Tile(1, 1) *whose supertile hierarchy covers the plane, with every tile carrying the decoration of Fig. 1, there is no closed strand: every strand is a two-way infinite path of ribbon pieces.*

Our second result is the single strand. We prove it for one specific tiling, the *nest tiling* $\mathcal{N}$, which is grown from a single tile N_0 by repeatedly taking the Δ-supertile N_{b+1} that has N_b as its child number 1 (Section 8). In the usual language of substitution tilings, $\mathcal{N}$ is the tiling generated from one tile as a seed by iterating the substitution about it, a fixed point of the substitution (in the combinatorial sense; compare [1]); in the tile addresses of [20, §2.8] (after [10]), N_0 has the address $\Delta \overset{1}{\leftarrow} \Delta \overset{1}{\leftarrow} \Delta \overset{1}{\leftarrow} \cdots$, a Δ at every level, each child 1 of the next. Figures 1, 2, 9 and 10 show blocks of it.

Theorem B (The Ariadne Theorem)**.** *In the nest tiling $\mathcal{N}$, with every tile carrying the decoration of Fig. 1, all the ribbon pieces of all the tiles form a single strand. It is a two-way infinite path that passes through every tile. More precisely, for every $b \geq 1$, all the ribbon arcs of the block N_b lie on a single arc of N_{b+4}, and of N_{b+3} when b is even.*

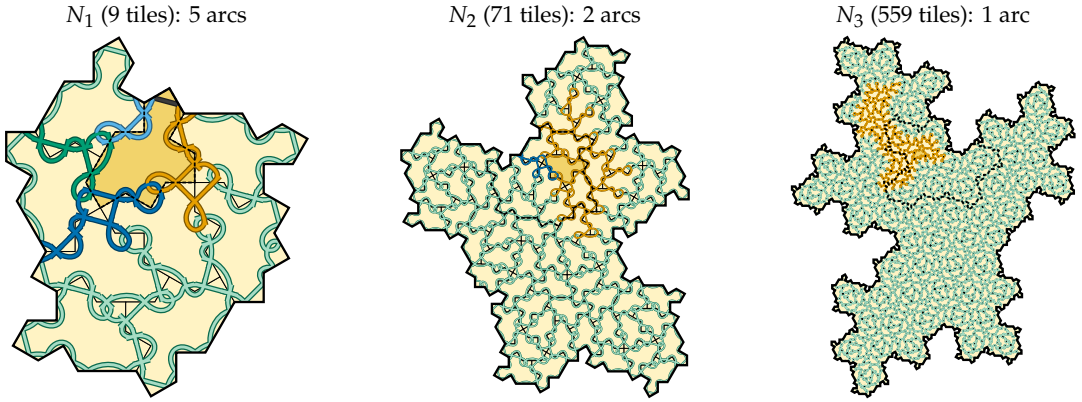


FIGURE 2. The arcs of the nest blocks N_1, N_2, N_3 that contain pieces of the original tile N_0 (shaded), one colour per arc. The dashed outline is the previous block.

Figure 2 shows the merging for N_0: its fourteen pieces lie on 5, 2 and 1 arcs of N_1, N_2 and N_3. Since the curves cross only transversally and in pairs, over and under can be assigned alternately, as in Celtic knotwork (Remark 3.6); the single strand then becomes an alternating diagram of a line.

The method. The proof has three ideas.

(1) *Black boxes.* Seen from outside, a supertile is just a list of which ribbon end on its boundary is joined to which. We show that every supertile of every kind and level has the same *signature*: a hook at every boundary vertex but two adjacent ones, A and B, where the two ribbons cross over; and no closed strand inside (Definition 4.3, Theorem 7.6).
(2) *Zigzags.* When two such boxes are glued along a common boundary path, the ribbons zigzag along it, and the outcome depends only on the parity of its length (Lemma 5.1). So lengthening a shared path by two edges changes nothing (Lemma 7.1).
(3) *Fixed parities.* The supertiles are combinatorially hexagons and compose by the substitution of [19, Fig. 5.1]. Their side lengths are given by Fibonacci numbers (Proposition 6.2); they only grow with the level, and their parities never change. So the side lengths at any level are reached from those at level 1 by lengthening, two edges at a time (Lemma 7.2), and a finite computation at level 1 gives the signature at every level.

The merging of all ribbons into one strand (Theorem 9.1) follows the same pattern. Where the ribbons of a block go one level up depends only on where they leave it, for most of its rim only on the side (Lemma 9.3); this map is computed at level 1 and carried to every level by the same lengthening. Followed three or four levels up, it puts every arc of a block on one strand.

Computer assistance. The proof uses computer checks of finitely many explicit facts: the signature of the Mystic and the level-1 data (vertex positions, side lengths, the position of A, and the signature of the level-1 supertiles), the agreement of our supertile data with [19, Figs. 4.1, 5.1], and two computations on the 18 compositions of level-1 supertiles, built from the tables and the level-1 side lengths alone: their signature (Computation 7.3) and, for Δ, the junction map (Computation 9.2). The geometric computations (the Mystic and level-1 data, and the composition pattern for $m \leq 3$) were carried out twice, by two separate programs: once in JavaScript with floating-point coordinates, and once in Python in exact arithmetic, with coordinates in $\mathbb{Z}[e^{i\pi/6}]$. The computations on the compositions are exact computations with integers, done by a Python program and again in the kernel of the Lean proof assistant. We also checked the conclusions directly on the actual supertiles up to level 9 (about 1.3×10^8

tiles), and by a second route, through skeletons and a symbolic computation with certified inequalities, which the proof does not use (Remarks 7.8 and 9.5). Appendix A lists every computation and the script that performs it. The programs, with logs of a full run, are in the ancillary files of this paper (folder `code/`).

Scope. Theorem A covers every tiling whose hierarchy covers the plane. Theorem B is proved for the nest tiling only, and we make no claim about the number of strands in any other tiling. The single strand is not a property of every such tiling: computations indicate that in at least one other tiling whose hierarchy covers the plane the pieces form exactly three strands (Remark 10.1). Tilings with faults are not covered by either theorem; see Section 10.

Related work. Curves drawn on tiles and joined across edges appear in several settings; in all of them each curve is closed or two-way infinite, as in Lemma 3.4. For Truchet tilings by squares carrying two quarter-circle arcs, the closed curves of random tilings have been studied by percolation and renormalisation methods [16, 12]. On rectangular grids, the mirror curves behind Celtic, Tchokwe and Tamil designs have a computable number of components; without internal mirrors it is the greatest common divisor of the side lengths [13]. On aperiodic tilings, the arc decoration of the Penrose rhombs due to Conway and Penrose gives closed curves apart from at most two that cross the whole tiling (see [9] and the references there), and Girih interlace drawn on Penrose tilings appears to consist of closed strands of unbounded variety [4]. Single curves through all the tiles of finite patches have been built as Hamiltonian cycles on Ammann–Beenker tilings [17], and as space-filling self-similar curves from decorations of aperiodic substitution tilings [15]. Closest to the present work, an online project [3] decorates selected edges of Tile$(1, 1)$ and describes, in one nested tiling, a single non-crossing curve that visits every tile once, with a machine check of its combinatorial core. That curve uses only some of the edges and has no crossings. Here every edge of every tile carries a ribbon end, the pieces cross, and the single strand is a knot diagram in the sense of Celtic knotwork. We are not aware of earlier work on such full knotwork decorations that form a single strand.

Use of AI and verification status. This work is an experiment in using AI for mathematical exploration. The author came up with the original idea of looking for a mono-stranded aperiodic knot, and then used an AI system (Claude, by Anthropic) to search for it; from the 34 candidates found, the author chose design 0.1:01 and drew its ribbons by hand. From then on the author and the AI system worked together: the AI system did the detailed work (the programs, the proof, the Lean formalisation and this paper), and the author directed, guided and corrected it. Lean checks the combinatorial core of the proof; the link with actual tilings is argued in words. The author has read parts of this paper and spot-checked figures and counts; most of the proof has not been reviewed by a human, and no professional mathematician has reviewed it. The AI system reviewed its own argument in several passes. A program or check called *independent* or *separate* here is a separate implementation by the same AI system. A fuller account: `https://cortado-j.github.io/SPECTRE_KNOT/made.html`.

Formal verification. The combinatorial part of the proof is formalised in the Lean 4 proof assistant [7], using its core library only (folder `lean/` of the ancillary files). Lean's kernel has checked the following, in a combinatorial form in which a patch, or a whole tiling, is described by which edge of which tile is glued to which edge of which other tile: every supertile of every kind and every level $m \geq 1$ has the signature (Theorem 7.6), and so contains no closed strand; in the union of any chain of supertiles, each a child of the next, in which every edge is eventually glued, every strand is a two-way infinite path (Theorem A), and the nest is such a chain; in the nest tiling, built as the union of the blocks N_b, the pieces form one two-way infinite strand (Theorem B), with the bounds of Theorem 9.1, which are exact (the last two rows of Table 5); and every vertex of every tile of N_b is surrounded by tiles

of N_{b+2} (Lemma 8.3(a), at the vertices of tiles). The kernel also re-runs the finite computations that these use (Computations 7.3 and 7.5, and the second part of Computation 9.2). The formal statements rest on Tables 1 to 3, as computed from the geometry, and on the nine level-1 supertiles as glued patches of tiles, computed in exact arithmetic by the program of Appendix A; Table 4 (read from [19, Fig. 4.2]) and the level-1 lengths ℓ_1, d_1 enter only the proofs, which check them against these. The link with tilings of the plane is not formalised; it is a stated assumption of the formal statements, in three parts: that the supertiles of an actual tiling are glued as these tables say, which rests on [19] and Section 2 (Theorem 2.1, Proposition 2.4 and Lemma 2.6); the first step of the proof of Theorem A, that a tiling whose hierarchy covers the plane is the union of such a chain; and the geometric Propositions 8.2 and 8.4, with the step of Lemma 8.3(a) from the vertices of tiles to the points of the plane. The Mystic (Lemma 4.4) is not treated separately: the formalisation starts from the level-1 supertiles. The formal proofs follow Sections 3 to 9, with these differences of detail. The strands of a tiling are modelled by two fixed-point-free involutions on ports (Lemma 3.4). Proposition 5.4 is proved for any gluing of boxes, with marks defined at the vertices of each child, so that the facts Section 5.2 reads off the angles at a vertex are proved from the gluing; (c) is proved in the form used, that if every reduced walk reaches the rim there is no closed strand. The formal proof of Lemma 7.1 follows the ribbon along the new path directly, and shows that a move creates no closed strand through Proposition 5.4(b),(c). The induction of Proposition 7.4 carries the signature in a form that finds each parent rim vertex by turning about a vertex of a child, and reads the adjacency of A and B off Table 3. In Computation 9.2 the kernel computes the second part; the first is proved for every level, at the level of tiles, from the signature of N_{m+1}; and in the proof of Lemma 9.3 the new vertices are followed port by port rather than by the last part of Lemma 5.1. Lemma 8.3(b) is proved from the tables, as its proof notes; the proof of Lemma 8.3(a) at the vertices of tiles uses that every supertile is a disc at its rim, which the paper takes from the geometry and the formalisation proves level by level, and it carries the table argument of (b) to vertices in place of the disjointness of sides 2 and 4 of N_{b+1}. The formalisation also proves facts the paper takes from the geometry: the gluing of every supertile is symmetric and glues no tile to itself, and Tables 2 and 3 agree with each other. Definition 5.3, Corollary 5.5 and the second routes of Remarks 7.8 and 9.5 are not formalised. The proofs use only Lean's standard axioms. The formalisation was written by the AI system as well; its statement file (`lean/SpectreKnot/Statement.lean`) contains every definition that the formal theorems depend on, with comments in plain English.

Outline. Section 2 recalls the facts about Spectre tilings that we use, in the notation of [19]. Section 3 defines the decoration and its strands. Section 4 introduces boxes and the signature. Section 5 proves the Zigzag Lemma and the reduction to skeletons. Section 6 computes the side lengths. Section 7 proves the signature at every level, by lengthening, and Theorem A. Section 8 constructs the nest tiling, and Section 9 shows that its arcs merge and proves Theorem B. Section 10 discusses scope and open questions. Appendix A lists the computations.

2. Spectre tilings

2.1. **The tile.** Let $\rho = e^{i\pi/6}$. The polygon Tile(1, 1) has fourteen edges of unit length; we label its vertices $v_0, \dots, v_{13}$ counter-clockwise and its edges $e_i = v_i v_{i+1}$ (indices mod 14), with $v_0 = 0$ and

$$v_{i+1} - v_i = \rho^{k_i}, \qquad (k_0, \dots, k_{13}) = (0, 10, 1, 3, 0, 2, 5, 7, 4, 6, 6, 8, 11, 9).$$

All vertices lie in $\mathbb{Z}[\rho]$, and the edge directions are multiples of 30°; see Fig. 5. This is the equilateral member Tile(1, 1) of the family Tile(a, b) [19, §§1–2]; as there, its long side is split into the two collinear edges e_9, e_{10}, so that v_{10} has interior angle 180°. Our Tile(1, 1) is the mirror image of the polygon drawn in [19, Fig. 1.1]; it has the handedness of the single

parent T	0	1	2	3	4	5	6	7
Γ	Π	Δ	Θ	Σ	Ξ	Φ	Γ	
Δ	Ξ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Θ	Ψ	Δ	Π	Φ	Σ	Π	Φ	Γ
Λ	Ψ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Ξ	Ψ	Δ	Π	Φ	Σ	Ψ	Φ	Γ
Π	Ψ	Δ	Ξ	Φ	Σ	Ψ	Φ	Γ
Σ	Ξ	Δ	Ξ	Φ	Σ	Π	Λ	Γ
Φ	Ψ	Δ	Ψ	Φ	Σ	Π	Φ	Γ
Ψ	Ψ	Δ	Ψ	Φ	Σ	Ψ	Φ	Γ
position in [19, Fig. 5.1]	L	D	B	R	S	TR	TL	G

TABLE 1. The children of each kind of supertile. For $T = \Gamma$ there is no child at position B, and the positions of children $2, \dots, 6$ are R, S, TR, TL, G. The positions name the places in the eight-hexagon layout of [19, Fig. 5.1]: L left, D the Δ, B bottom, R right, S the Σ, TR top right, TL top left, G the Γ.

Spectre and of the Mystic drawn in [19, Fig. 2.1]. *Spectres* are tiles, obtained for instance by replacing every edge of Tile(1, 1) by a suitable curve, whose tilings correspond exactly to the homochiral tilings by Tile(1, 1) [19, §2, Lemma 2.1]. We work throughout with Tile(1, 1) and *homochiral* tilings, in which any two tiles are related by a rotation and a translation; reflecting the whole plane (with the decoration below, which maps strands to strands) if necessary, we assume that every tile is the image of Tile(1, 1) under a rotation and a translation. Since the correspondence keeps the isometries g_i [19, §2], and [19, p. 5] treats the two as the same universe of tilings, what [19] proves for tilings by Spectres holds for homochiral tilings by Tile(1, 1), and we use it in that form. Such tilings exist [19, Thm. 2.2], and every tiling by Tile(1, 1) is vertex to vertex, so tiles meet full edge to full edge [19, proof of Lemma 2.1]. The tile has no non-trivial symmetry, proper or improper (its cyclic sequence of edge directions has none), so an isometry between two tiles is unique.

2.2. **Supertiles.** We use the hierarchy of [19]. There are nine *kinds*

$$\mathcal{K} = \{\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi\}.$$

A *level*-0 *supertile* of kind $K \neq \Gamma$ is a single tile; a level-0 supertile of kind Γ is a *Mystic* [19, Fig. 2.1], the pair of tiles $\{\text{Tile}(1,1),\ g(\text{Tile}(1,1))\}$ with $g(z) = \rho z + v_8$, and its congruent copies; we call Tile(1, 1) its *first* tile. (The outline of the Mystic, which [19] calls symmetric, has a mirror symmetry, but the Mystic has no non-trivial proper symmetry: a proper symmetry exchanging its tiles would be an involution taking Tile(1, 1) to $g(\text{Tile}(1,1))$, but the only proper motion doing so is g, a rotation by 30°. So the first tile is determined by the pair.) A level-$(m+1)$ supertile is the union of level-m supertiles, its *children*, arranged in a fixed way: eight children, or seven for kind Γ, of the kinds listed in Table 1. A level-1 supertile has the kind of its cluster in [19, Fig. 4.1], which is determined by the positions of its six vertices (Section 2.3) [19, proof of Thm. 4.1]; a level-$(m+1)$ supertile, $m \geq 1$, has kind T if its children form the panel T of [19, Fig. 5.1]. For $m \geq 1$, child c is the child at the position in the last row of Table 1; at level 1 the children are numbered as in Fig. 3. (This is the order used by Kaplan's implementation of the key-point construction of [19, App. A], see [14]. That implementation gives Table 1 and, tile for tile, the same supertiles as `exact/exact.py` up to level 4; see `crosscheck/kaplan_compare.py`.) For definiteness a single tile gets the kind of its position in its level-1 supertile; only the distinction between Γ and the other kinds is used at level 0. In particular child 1 of every supertile is a Δ, and the last child is a Γ; at levels $m \geq 1$ a Δ occurs only as child 1.

The nine level-1 supertiles are the Spectre clusters (9 tiles) and the Mystic cluster (8 tiles, kind Γ) of [19, Fig. 2.1], with the markings of [19, Fig. 4.1], up to reflection: [19, Fig. 2.1] draws

the clusters with tiles of the opposite handedness to its single Spectre, and so to Tile(1, 1). They are drawn, with tiles of the handedness of Tile(1, 1), in Fig. 3. We number levels so that our level-m supertiles are made of level-$(m-1)$ supertiles. In the language of [19], the level-1 supertiles are the clusters, which are combinatorially the marked hexagons of [19, Fig. 4.2], and the level-n supertiles of hexagons of [19, §5] are our level-$(n+1)$ supertiles.

The following theorem restates, in our language, results of [19].

Theorem 2.1 (Smith, Myers, Kaplan, Goodman-Strauss [19])**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile(1, 1)*.*

(a) [19, Thms. 2.2, 3.1, 4.1, §5, Fig. 2.1] *The tiles of $\mathcal{T}$ group uniquely into level-1 supertiles. Each contains one Mystic, in the position shown in* [19, Fig. 2.1]*; this Mystic and the other tiles, taken singly, are the level-0 supertiles inside it. For every $m \geq 1$ the level-m supertiles of $\mathcal{T}$ group uniquely into level-$(m+1)$ supertiles. Each tile lies in a unique nested sequence $X_0 \subset X_1 \subset \cdots$ with X_m a level-m supertile.*

(b) [19, Thms. 3.1, 4.1, 5.1, §5] *For every $m \geq 1$, the tiling of the plane by the level-m supertiles of $\mathcal{T}$ is combinatorially equivalent (homeomorphic as a topological complex* [19, §2.2]*) to a tiling by regular hexagons, each level-m supertile of kind K corresponding to a hexagon marked as the hexagon K of* [19, Fig. 4.2] *for even m, and as its mirror image for odd m, adjacent supertiles corresponding to hexagons sharing an edge with matching labels; and the grouping into level-$(m+1)$ supertiles corresponds, up to reflection, to the grouping of hexagons shown in* [19, Fig. 5.1].

In (a), [19, Thm. 4.1] gives the grouping into clusters; its uniqueness, like that of the higher levels, is part of [19, Thm. 2.2], and we use it only to speak of *the* hierarchy of $\mathcal{T}$. In (b), the handedness is that of our level-1 supertiles, which are the mirror images of the clusters drawn in [19, Figs. 2.1, 4.1] (see the base case of $(*)$ in the proof of Proposition 2.4); it alternates with m by [19, Thm. 5.1].

In particular every level-m supertile ($m \geq 1$) is a closed topological disc whose boundary, its *rim*, is a simple closed polygon made of tile edges. Its *vertices* are the six points of the rim where it meets two other level-m supertiles, and its *sides* are the six arcs of the rim between consecutive vertices. (The tile vertices on a rim are called *rim vertices*; for a supertile of level ≥ 1, "vertex" alone always means one of the six.) A vertex of one supertile that lies on the rim of another is a vertex of that one too, since exactly three supertiles meet there. Each side is shared, as a path of tile edges, with exactly one neighbouring supertile.

2.3. **Own orientation, side numbering and the crossover corner.** By [19, Thm. 5.1], the supertiles composed from a tiling by marked hexagons are combinatorially equivalent to the *mirror images* of those hexagons. So, at each level, the cyclic order of the labelled sides is reversed, and it is convenient to orient supertiles alternately.

Definition 2.2 (Own orientation)**.** The *own orientation* of a level-m supertile is counterclockwise if m is even and clockwise if m is odd. We write $\omega_m = (-1)^m$.

Thus a child and its parent are always oriented oppositely. Each supertile has a distinguished rim vertex A, its *crossover corner*, defined recursively: for a tile, $A = v_0$; for a Mystic, A is the vertex v_0 of its first tile Tile(1, 1); for a level-$(m+1)$ supertile, A is the crossover corner of its Γ child. (That this point lies on the rim is computed for levels 0 and 1 and follows from Proposition 2.4(d) at higher levels.) The second corner B, adjacent to A, appears in Theorem 7.6; we call A and B the *crossover corners*. The vertices of a level-m supertile ($m \geq 1$) are numbered $0, \ldots, 5$ in own order, starting with the first vertex at or after A; side j runs from vertex j to vertex $j+1$. Figure 3 shows this numbering for the level-1 supertiles.

2.4. **The composition pattern.** Theorem 2.1(b) says that every level of the hierarchy is assembled in the same combinatorial way. We record this in our numbering.

Definition 2.3 (Composition pattern)**.** The *composition pattern* C consists of the following data for each parent kind T, given in Tables 1 to 3 and drawn in Fig. 4.

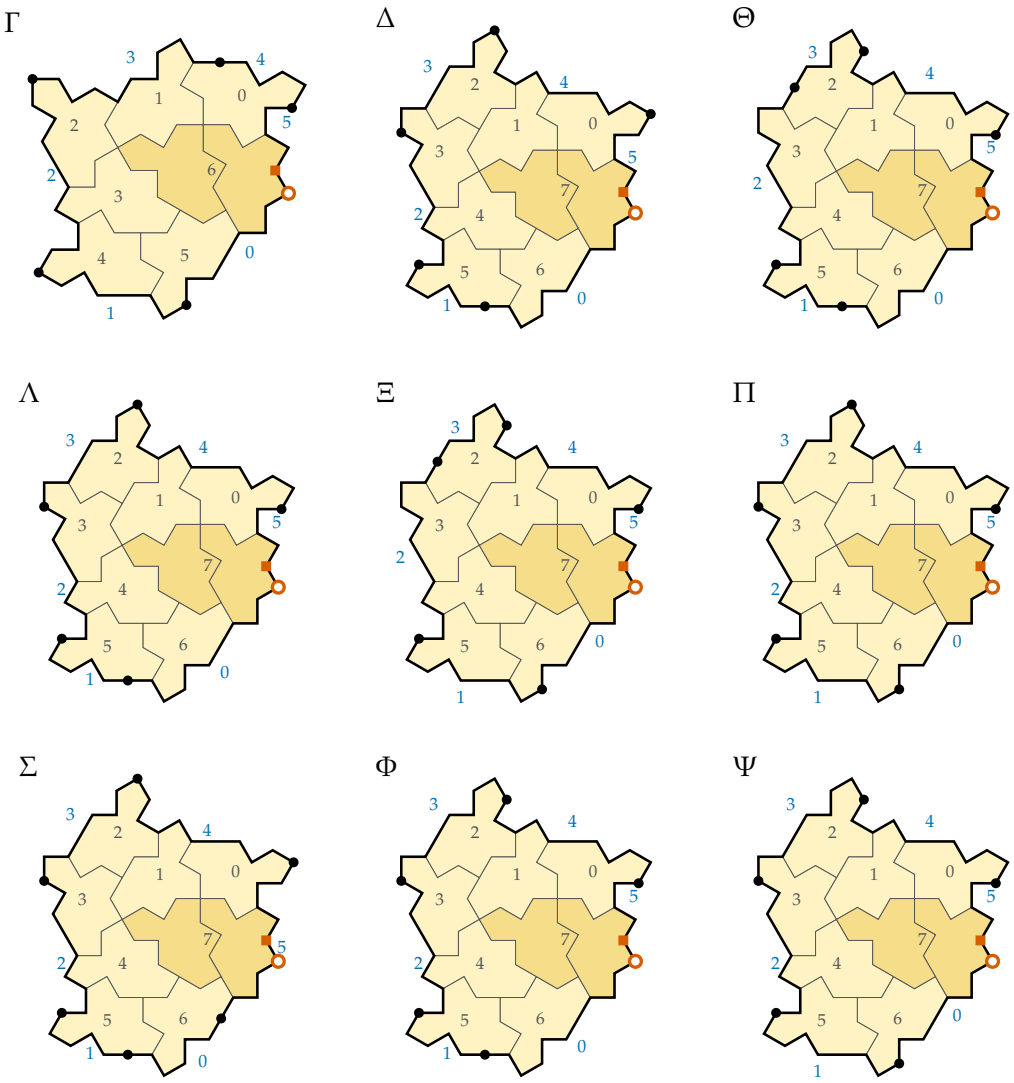


FIGURE 3. The nine level-1 supertiles, with true handedness. The Mystic is the darker pair of tiles. Grey numbers: the children (Table 1); black dots: the six vertices; blue numbers: the sides, numbered clockwise (the own orientation at level 1); open circle: the crossover corner A, which is vertex 0 except for Σ; filled square: the corner B (Theorem 7.6). For Σ, A lies inside side 5.

(i) The kinds $K_0, \ldots, K_{n_T-1}$ of the children (Table 1).
(ii) For each child c and each side j of c, either the sibling c' and its side j' that lie across side j, or the symbol ∂ if side j lies on the parent's rim (Table 2).
(iii) For each parent side J, the sequence of child sides (c, j) that make up side J, in the parent's own order (Table 3).

Proposition 2.4 (The pattern holds at every level)**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile$(1,1)$*, let $m \geq 1$, and let Y be a level-$(m+1)$ supertile of $\mathcal{T}$ of kind T, with children $X_0, \ldots, X_{n_T-1}$. Then:*

(a) *X_c has kind K_c (Table 1);*
(b) *side j of X_c is the same path of tile edges as side j' of $X_{c'}$, traversed in the opposite direction, whenever Table 2 lists $c'.j'$, and it lies on the rim of Y when the table lists ∂;*
(c) *side J of Y is the concatenation of the children's sides listed in Table 3, and vertex J of Y is the vertex $j+1$ of the child c whose side $c.j$ is listed first;*

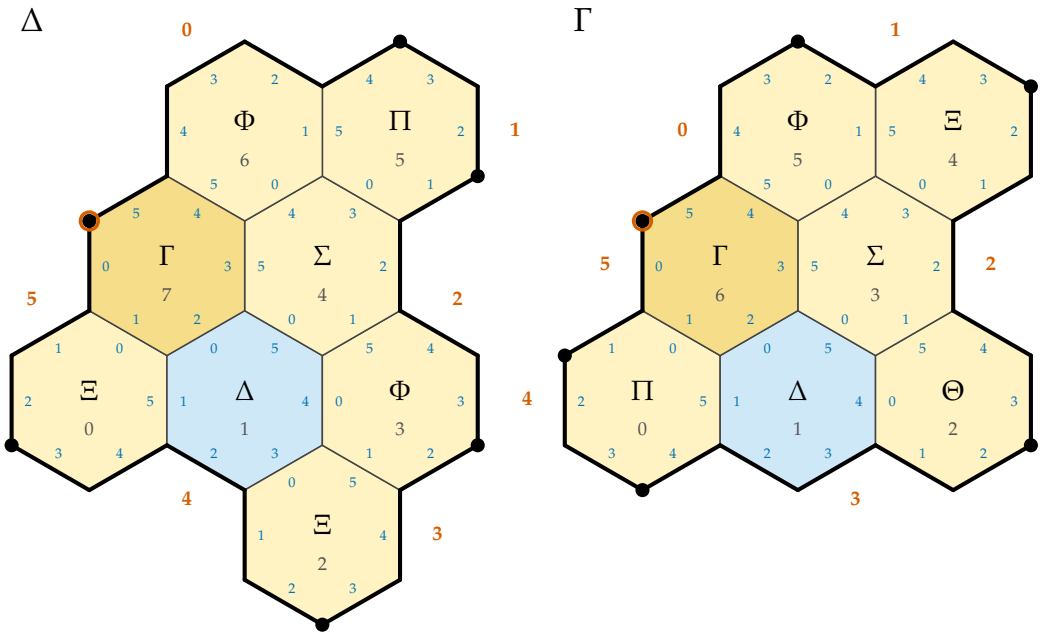


FIGURE 4. The composition pattern C for Δ (left; the same for all kinds except Γ, with the children's kinds as in Table 1) and for Γ (right), in the layout of [19, Fig. 5.1]. Each hexagon is a child: its kind, its number (grey), and its sides 0–5 (blue). Thick black: the parent's rim; black dots: the parent's vertices; vermilion numbers: the parent's sides; circled: the parent's crossover corner, which is vertex 0 of the Γ child. Children's sides are numbered in one rotational sense and the parent's in the other.

child	side 0	side 1	side 2	side 3	side 4	side 5
0	7.1	∂	∂	∂	∂	1.1
1	7.2	0.5	∂	2.0	3.0	4.0
2	1.3	∂	∂	∂	∂	3.1
3	1.4	2.5	∂	∂	∂	4.1
4	1.5	3.5	∂	5.0	6.0	7.3
5	4.3	∂	∂	∂	∂	6.1
6	4.4	5.5	∂	∂	∂	7.4
7	∂	0.0	1.0	4.5	6.5	∂

child	side 0	side 1	side 2	side 3	side 4	side 5
0	6.1	∂	∂	∂	∂	1.1
1	6.2	0.5	∂	∂	2.0	3.0
2	1.4	∂	∂	∂	∂	3.1
3	1.5	2.5	∂	4.0	5.0	6.3
4	3.3	∂	∂	∂	∂	5.1
5	3.4	4.5	∂	∂	∂	6.4
6	∂	0.0	1.0	3.5	5.5	∂

TABLE 2. Adjacency of the children's sides: the entry $c'.j'$ in row c, column j means that side j of child c is side j' of child c'; ∂ means that it lies on the parent's rim. Upper table: every parent kind except Γ. Lower table: Γ.

(d) *the crossover corner A of Y is vertex* 0 *of its* Γ *child, which is vertex* 0 *of Y if $T \neq \Sigma$, and lies inside side* 5 *of Y if $T = \Sigma$.*

Eight children (every $T \neq \Gamma$): the parent's rim, in own order from vertex 0 of child 7.

position	0	1	2	3	4	5	6	7	8	9	10
piece	7.5	6.4	6.3	6.2	5.4	5.3	5.2	5.1	4.2	3.4	3.3
position	11	12	13	14	15	16	17	18	19	20	21
piece	3.2	2.4	2.3	2.2	2.1	1.2	0.4	0.3	0.2	0.1	7.0

Seven children ($T = \Gamma$), from vertex 0 of child 6; sides $0, \dots, 5$ begin at positions $0, 3, 6, 11, 16, 18$.

position	0	1	2	3	4	5	6	7	8	9
piece	6.5	5.4	5.3	5.2	4.4	4.3	4.2	4.1	3.2	2.4
position	10	11	12	13	14	15	16	17	18	19
piece	2.3	2.2	2.1	1.3	1.2	0.4	0.3	0.2	0.1	6.0

parent	first position of side					
	0	1	2	3	4	5
Δ	0	5	7	11	14	19
Θ	0	5	7	13	15	20
Λ	0	5	7	11	14	20
Ξ	0	3	7	13	15	20
Π	0	3	7	11	14	20
Σ	2	5	7	11	14	19
Φ	0	5	7	11	15	20
Ψ	0	3	7	11	15	20

TABLE 3. The parent's sides as concatenations of children's sides. A piece $c.j$ is side j of child c, which the parent's own orientation traverses from vertex $j+1$ to vertex j of the child. Side J of the parent consists of the pieces from its first position up to the first position of side $J+1$ (cyclically). For $T \neq \Sigma$ the parent's side 0 begins at vertex 0 of the Γ child, which is the parent's crossover corner. For $T = \Sigma$ the crossover corner lies inside side 5 (positions 19–1), after the pieces 0.2, 0.1, 7.0.

Justification. By Theorem 2.1(b), the level-m supertiles inside Y are combinatorially the hexagons of the corresponding panel of [19, Fig. 5.1]: hexagon edges correspond to sides and hexagon vertices to vertices. We use the following dictionary between the labels of [19, Fig. 4.2] and our side numbers:

(∗) side j of a level-m supertile of kind K carries the label of the edge of hexagon K in direction W, SW, SE, E, NE, NW for $j = 0, \dots, 5$ respectively (Table 4); its crossover corner A is vertex 0 if $K \neq \Sigma$; if $K = \Sigma$, A is not a vertex, so it lies in the relative interior of side 5.

We show (∗) by induction on m. For $m = 1$ it is a computation on the level-1 supertiles, which have $E_1 = 44$ rim edges for Γ and 46 for the other kinds; their vertices lie at the rim positions, counted from A in own order,

$$\begin{array}{lll} \Gamma : 0, 7, 14, 24, 34, 39 & \Delta : 0, 10, 15, 23, 30, 40 & \Theta : 0, 10, 15, 26, 31, 41 \\ \Lambda : 0, 10, 15, 23, 30, 41 & \Xi : 0, 7, 15, 26, 31, 41 & \Pi : 0, 7, 15, 23, 30, 41 \\ \Sigma : 4, 10, 15, 23, 30, 40 & \Phi : 0, 10, 15, 23, 31, 41 & \Psi : 0, 7, 15, 23, 31, 41. \end{array}$$

Drawn with the edges in odd directions lengthened to $\sqrt{3}$, which turns the tiles into hats and turtles [19, §3], and in the handedness of [19, Fig. 4.1], the nine level-1 supertiles coincide with the nine marked clusters there: the same outlines, the vertices at the marked dots, and the labels of Table 4 in the places where that figure has them; A is the dot between the sides labelled W and NW (for Σ, a point of the NW side), on the light tile of the Mystic. A script checks this against a render of the figure (Appendix A): for each cluster it finds the six dots, fits our six vertices to them by a proper similarity with residual below 1% of an edge (any other matching, or the vertices of any other kind, fits at least 40 times worse), finds our whole rim on the printed outline and our first Mystic tile on the lighter shaded tile, and compares the labels, read from the figure by hand, with Table 4. Suppose (∗) holds for the children of Y. The panel does not label the inner hexagons, but the matching rules fix the rotation of each of them, and there is exactly one consistent choice (it is found by the script

of Remark 2.5). With this dictionary, Tables 2 and 3 are then a transcription of the panel of [19, Fig. 5.1] for T, including its vertices (the six marked points on its outline); this gives (a)–(c). The crossover corner of Y is that of its Γ child, which is the child's vertex 0 by (∗), and Table 3 places this vertex at the start of the parent's side 0 if $T \neq \Sigma$ and inside side 5 if $T = \Sigma$; this is (d). Finally, the labels that the panel assigns to the parent's sides, numbered from this point in the parent's own order, are those of Table 4 for every T. So (∗) holds for Y. In particular A always lies on the rim. □

Remark 2.5 (Computer cross-checks of the pattern). Since Tables 2 and 3 carry the whole weight of the citation, we checked them in two independent ways. First, we built the actual supertiles of levels 1 to 8 with the key-point construction of [19, App. A] and extracted the pattern of each composition $m \to m+1$ for $m = 1, \dots, 7$. For $m \leq 4$ the vertices were found as the points where three level-m supertiles meet, observed inside larger supertiles. For $m = 5, \dots, 8$ they were propagated by the rule of Proposition 2.4(c), learnt at the observed level of the same parity; at levels 5 and 6 the propagated vertices agree with the observed ones (for Ψ, with the four of its six vertices that were observed). So for $m = 6, 7$ the rule (c) is an input of the extraction rather than something it tests. All seven extracted patterns are identical to each other and to Tables 2 and 3; for $m \leq 3$ this was done in exact arithmetic (Appendix A). Second, the nine panels of [19, Fig. 5.1] were encoded: kinds, positions and edge labels by hand, and the marked vertices (with the centre of each panel's Γ hexagon) by a script from a render of the figure, which re-runs from the PDF. A script then confirmed, for all nine kinds, that the encoding agrees with Tables 2 and 3, and that the side-to-label dictionary above is the same in all 71 placements of a child. Two deliberately corrupted encodings (two outer labels exchanged; one wrong label in [19, Fig. 4.2]) were rejected. We also checked that the tiles of every supertile of levels 1 to 7 (up to 2,147,679 tiles) have disjoint interiors and exactly fill its rim.

We shall use one consequence of Theorem 2.1(a) and Proposition 2.4 repeatedly.

Lemma 2.6 (Rigidity)**.** *Any two level-m supertiles of the same kind, in possibly different homochiral tilings, are related by a rotation and translation that maps descendants to descendants with their numbers (children, their children, and so on, down to the tiles), and it is the only such motion. For $m \geq 1$ it maps the six vertices, the sides and the corner A of one onto those of the other.*

Proof. Induction on m. For $m = 0$ this is clear, since the Mystic is a fixed pair of tiles without proper symmetry. For $m = 1$, the level-1 supertiles of a homochiral tiling are congruent copies of the clusters of Fig. 3 [19, Fig. 2.1, §4], with their vertices at the points marked in [19, Fig. 4.1]; the numbers of their children are those of the reference clusters of Fig. 3, and A is fixed by the Mystic. Let $m \geq 1$. By induction, the children of two level-$(m+1)$ supertiles Y, Y' of kind T are pairwise related by unique proper motions g_c, which map the vertices of child c of Y onto those of child c of Y' with their numbers. If side j of child c is glued to side j' of child c', then by Proposition 2.4(b), applied to Y and to Y', vertices j and $j+1$ of c coincide with vertices $j'+1$ and j' of c', in Y and in Y'. So g_c and $g_{c'}$ agree on two distinct points; being proper motions, they are equal. The children's adjacency graph is connected, so all the g_c are one motion g. By Proposition 2.4(c),(d), the vertices and the corner A of Y are vertices and the corner A of children, so g maps them, and hence the sides, onto those of Y'. Since g agrees with each g_c, it maps descendants to descendants with their numbers. Any proper motion that does so maps the first tile of the Mystic at the bottom of the chain of last (Γ) children to its counterpart, so it equals g, since a tile has no symmetry. □

Write $E_m(K)$ for the number of rim edges (equally, rim vertices) of a level-m supertile of kind K, and $\ell_m(K, j)$ for the number of edges of its side j. By Lemma 2.6 these numbers are well defined.

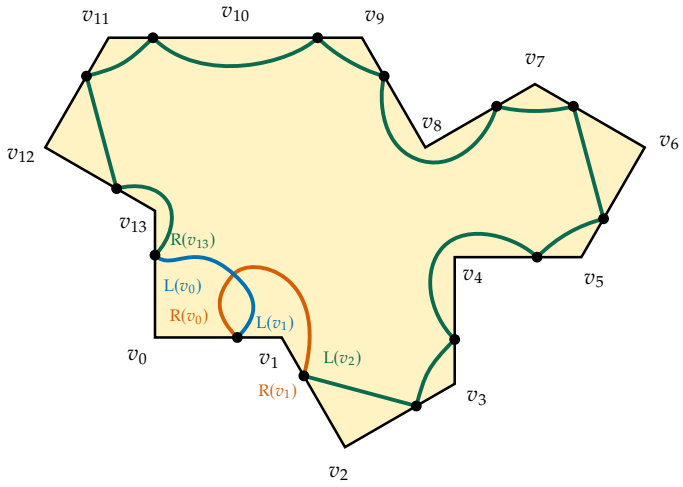


FIGURE 5. The decoration on the true geometry (centre lines of the ribbons of Fig. 1). Each edge carries one crossing point (dot), at which two pieces of the tile end: one leaves towards each end of the edge. On e_i these two ends are the ports $\mathrm{R}(v_i)$ (towards v_i) and $\mathrm{L}(v_{i+1})$ (towards v_{i+1}); the labels show them round the crossover corners. Green: the twelve hooks $\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i)$. Vermilion: $\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1)$; blue: $\mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)$, the crossover at the corners $A = v_0$ and $B = v_1$.

3. The decoration and its strands

3.1. **Ports and pieces.** Let t be a tile and e one of its edges, with endpoints x and y. The tile has two *ports* on e, the port *near* x and the port *near* y; formally, they are the pairs (e, x) and (e, y). In the drawing (Fig. 5), e carries one crossing point, and the two ribbon ends of t that meet it, one running off towards x and one towards y, are these two ports. For a vertex v_i of the tile, write

$$\mathrm{R}(v_i) = \text{the port on } e_i \text{ near } v_i, \qquad \mathrm{L}(v_i) = \text{the port on } e_{i-1} \text{ near } v_i,$$

the ports just after and just before the corner v_i in counter-clockwise order. Each of the 28 ports is $\mathrm{R}(v_i)$ or $\mathrm{L}(v_i)$ for exactly one i.

Definition 3.1 (The decoration)**.** The *decoration* joins the 28 ports of Tile$(1,1)$ in pairs by fourteen *pieces*:

$$\underbrace{\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i) \quad (2 \le i \le 13)}_{\text{twelve hooks}}, \qquad \underbrace{\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1), \qquad \mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)}_{\text{the crossover}}.$$

Every tile of a homochiral tiling carries the image of this decoration.

In our drawing the two crossover pieces cross exactly once, transversally, and this is the only crossing inside the tile (Remark 3.6). Each hook turns back round one vertex. Figure 5 shows the pieces and their ports, and Fig. 1 the ribbons as drawn. The drawing uses cubic Bézier curves, with the crossing point of edge e_i at distance $0.5 + 0.15(-1)^i$ from v_i. Only the pairing of Definition 3.1 matters for the theorems; any drawing that realises it may be used.

3.2. **The straight-through rule.** Let $\mathcal{T}$ be a homochiral tiling, and let e be an edge shared by tiles t and t', with endpoints x and y.

Definition 3.2 (Straight-through rule)**.** The port of t on e near x is *joined* to the port of t' on e near y, and the port of t near y to the port of t' near x.

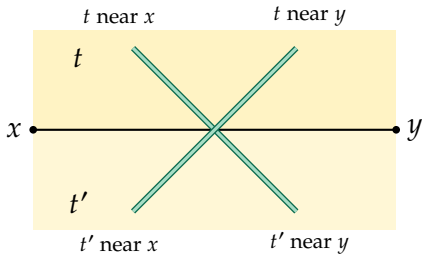


FIGURE 6. The straight-through rule at the crossing point of an edge xy: the end of t heading towards x continues as the end of t' heading towards y.

This is what a ribbon does when it continues straight through the crossing point (Fig. 6). In the drawing the two tiles' crossing points on e coincide. The reason is that tiles in a homochiral tiling traverse a shared edge in opposite directions, and an edge e_i of one tile is always glued to an edge e_j of the other with $i + j$ odd; this follows from the vertex colouring of [19, proof of Lemma 2.1]: the interior angles of Tile$(1, 1)$ alternate between multiples of 90° and of 120°, and these classes must agree where vertices meet. As v_i of one tile meets v_{j+1} of the other, i and $j+1$ have the same parity. We verified it on all supertiles up to level 4. We also checked that the drawn ribbons realise the pairing on all supertiles up to level 3 (Appendix A). The proofs use only Definitions 3.1 and 3.2.

3.3. **Strands.**

Definition 3.3 (Strand graph)**.** Let $\mathcal{T}$ be a homochiral tiling. The *strand graph* $\mathcal{G}(\mathcal{T})$ is the multigraph with one vertex for each piece of each tile and one edge for each pair of ports joined by the straight-through rule, between the two pieces having these ports as ends. A *strand* is a connected component of $\mathcal{G}(\mathcal{T})$; a strand is *closed* if it is finite.

Lemma 3.4. *Every vertex of $\mathcal{G}(\mathcal{T})$ has degree* 2*. Hence every strand is either a finite cycle (a closed strand) or a two-way infinite path.*

Proof. A piece has two ends, each a port of its tile on some edge e. In a tiling of the plane, e is shared by exactly one other tile, so each end is joined to exactly one port, an end of a piece of that neighbour. So every vertex has degree exactly 2; joins always link pieces of different tiles, so $\mathcal{G}(\mathcal{T})$ has no loops. A connected, locally finite multigraph with all degrees 2 is a finite cycle (possibly of length 2) or a two-way infinite path. □

Remark 3.5 (A single strand needs a non-periodic tiling). Let $\mathcal{T}'$ be any edge-to-edge tiling of the plane by compact decorated tiles (not necessarily copies of Tile$(1, 1)$), each carrying at least one piece, with one crossing point on each edge, and with strands defined as above by the straight-through rule. Suppose that $\mathcal{T}'$, decoration included, is invariant under a translation $\tau \neq 0$, and that its strand graph is a single two-way infinite path, with vertices $x \in \mathbb{Z}$. Then τ acts on the path by a graph automorphism, and freely, since a non-zero translation moves every tile. An automorphism of the path is $x \mapsto x + k$ or $x \mapsto c - x$. A reflection $x \mapsto c - x$ satisfies $\tau^2 = \mathrm{id}$ on the path, but τ^2 is again a non-zero translation and fixes no piece; so τ is $x \mapsto x + k$ with $k \neq 0$, and the pieces fall into $|k|$ orbits under $\langle\tau\rangle$. But the tiles fall into infinitely many orbits: the images of one tile from each orbit cover the cylinder $\mathbb{R}^2/\langle\tau\rangle$, which has infinite area. Since every tile has a piece, the pieces also fall into infinitely many orbits. This contradiction shows that the ribbons of a decorated tiling with a translational symmetry never form a single strand. Every tiling by Tile$(1, 1)$ of one handedness is non-periodic [19, Thm. 2.2], so this obstruction does not arise there.

Remark 3.6 (Over and under). Draw the pieces of every tile as in Fig. 1. We checked numerically that each piece stays inside its tile away from its ends, that the two crossover pieces cross exactly once, transversally (at about 90°), and that no other two pieces of a tile

meet except at their common ends on the edges; and that at each crossing point the two pieces of a tile meet the edge at ±45° (Appendix A). So a ribbon continues smoothly across an edge, and the curves in the plane cross only transversally, in pairs: once inside each tile and at right angles at each edge. Colour each region of the complement of the curves by the parity of the number of crossings of a generic path from a fixed region. This is well defined: the parity of the number of crossings of a generic closed loop does not change when the loop is moved across an arc of the curves, or across a crossing point, where four arcs meet; and every closed loop in the plane can be shrunk to a point. Adjacent regions get different colours. The usual checkerboard rule then makes over and under alternate along every strand [6]. With such a choice the single strand of Theorem B becomes an alternating diagram of a line, as in Celtic knotwork. None of this affects which pieces lie on which strand.

4. Boxes and the signature

4.1. **Patches and boxes.** A *patch* is a finite set P of tiles of a homochiral tiling whose union is a closed disc with a simple polygonal boundary, the *rim* ∂P. Every supertile of level ≥ 1 is a patch (Theorem 2.1), and so are a tile and a Mystic (whose two tiles share four consecutive edges, $e_6, \dots, e_9$ of the first tile; its rim has 20 edges). Every rim edge belongs to exactly one tile of P. Let $E = E(P)$ be the number of rim edges, and number the rim vertices $u_0, \dots, u_{E-1}$ in a chosen orientation of the rim, indices mod E. For a supertile we always use its own orientation (Definition 2.2). For a rim vertex u_k define the *rim ports*

$$\mathrm{R}(u_k) = \text{the port on } u_k u_{k+1} \text{ near } u_k, \qquad \mathrm{L}(u_k) = \text{the port on } u_{k-1} u_k \text{ near } u_k,$$

where the port belongs to the tile of P containing that edge. There are $2E$ rim ports. For the tile itself, with its counter-clockwise orientation, these are the ports $\mathrm{R}(v_i), \mathrm{L}(v_i)$ of Section 3. Reversing the orientation exchanges R and L at every vertex.

Lemma 4.1 (The box of a patch)**.** *Start at a rim port of P and follow the pieces of the tiles of P, crossing every interior edge of P by the straight-through rule. The path ends at a different rim port. This defines an involution π_P on the rim ports without fixed points. The pieces of P that are not reached from any rim port form finitely many cycles, the* closed strands of P*; write $\kappa(P)$ for their number.*

Proof. The pieces of P and the joins across interior edges form a finite graph in which every piece has degree 2, except that a piece end at a rim port has no partner. Its components are therefore paths, whose two ends are rim ports, and cycles. Paths have two distinct ends. □

We call the paths of Lemma 4.1 the *arcs* of P; there are exactly E of them. The triple $(\partial P, \pi_P, \kappa(P))$ is the *box* of P. Everything we need about the inside of P is in its box, as the next lemma shows.

Lemma 4.2 (Gluing)**.** *Let a patch P be the union of patches $P_1, \dots, P_n$ with pairwise disjoint interiors. Then the box of P is determined by the boxes of the P_i together with the identification of their shared rim edges. More precisely, π_P is computed by alternately applying the π_{P_i} and the straight-through rule across shared edges, until a rim port of P is reached. The closed strands of P are those of the P_i together with the cycles of this alternation that contain no rim port of P.*

Proof. Every arc or closed strand of P (Lemma 4.1) is a concatenation of arcs and closed strands of the P_i, joined across edges shared by two of the P_i. □

4.2. **The signature.**

Definition 4.3 (Signature)**.** Let A and B be adjacent rim vertices of a patch P. We say that P has the *signature at* (A, B) if

(i) $\pi_P(\mathrm{L}(u)) = \mathrm{R}(u)$ for every rim vertex $u \notin \{A, B\}$ (a *hook* at u);
(ii) $\pi_P(\mathrm{R}(A)) = \mathrm{R}(B)$ and $\pi_P(\mathrm{L}(A)) = \mathrm{L}(B)$ (the *crossover*);

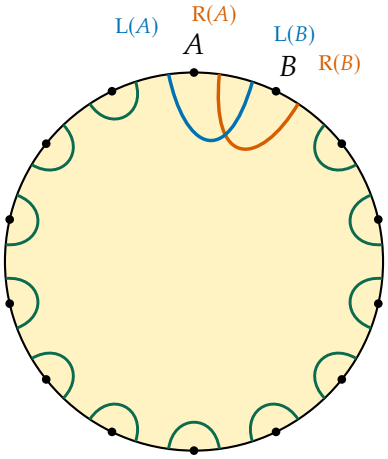


FIGURE 7. The signature at (A, B), drawn for a rim with 14 vertices oriented clockwise: a hook at every rim vertex except A and B, and the two crossover arcs $\mathrm{R}(A) \leftrightarrow \mathrm{R}(B)$ and $\mathrm{L}(A) \leftrightarrow \mathrm{L}(B)$.

(iii) $\kappa(P) = 0$.

The definition is symmetric in A and B and does not depend on the orientation of the rim, since reversing it exchanges L and R. A patch with the signature has $E - 2$ hook arcs, one at each rim vertex other than A and B, and two crossover arcs. Figure 7 is a picture. By Definition 3.1:

Lemma 4.4. *A tile has the signature at (v_0, v_1), and a Mystic has the signature at the vertices v_0, v_1 of its first tile.*

Proof. For the tile this is Definition 3.1. For the Mystic it is a finite computation with Lemma 4.2: there is no closed strand, the 18 rim vertices other than these two are hooks, and the crossover persists (Appendix A). □

5. The Zigzag Lemma and the skeleton

This section studies how the ribbons of glued boxes join. Section 5.1 proves the Zigzag Lemma, on which the proofs in Sections 7 and 9 rest. Section 5.2 condenses a composition to its *skeleton* and follows ribbons on it (Proposition 5.4); the Lean formalisation uses Proposition 5.4, and the second routes of Remarks 7.8 and 9.5 use the skeleton.

Throughout this section a patch P is the union of patches (*children*) $X_0, \dots, X_{n-1}$ with disjoint interiors, all of them oriented in the same rotational sense, and P is oriented in that sense too. Each child X_c has the signature at (A_c, B_c). When P is a supertile and the X_c are its children, the children carry their own orientation, so P is oriented opposite to its own orientation. This does not matter for the signature, which does not depend on the orientation (Section 4); but it does exchange L and R at the parent's rim vertices.

5.1. Runs and the Zigzag Lemma.

Two children $c \neq c'$ oriented in the same sense traverse a common rim path in opposite directions. For a rim vertex u of X_c we write $\mathrm{R}_c(u), \mathrm{L}_c(u)$ for the rim ports of X_c in the common orientation fixed above (for supertiles, their own orientation).

Lemma 5.1 (Zigzag Lemma)**.** *Let $u_0 u_1 \cdots u_n$ ($n \geq 1$) be a path of n edges lying on the rims of both X_c and $X_{c'}$, traversed by X_c from u_0 to u_n. Suppose that none of the interior vertices $u_1, \dots, u_{n-1}$ is a crossover corner of X_c or of $X_{c'}$. Then, alternately applying the straight-through rule across the*

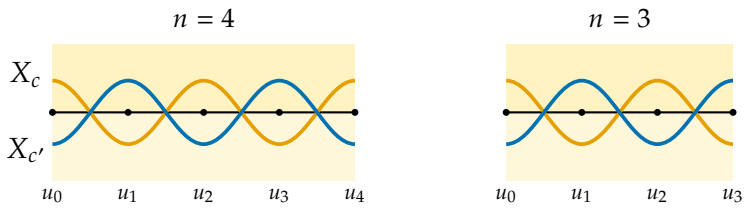


FIGURE 8. The Zigzag Lemma. Along a shared path whose interior corners are hooks on both sides, the ribbons cross the path at every edge and turn back at every corner, alternately in X_c (above) and $X_{c'}$ (below). For even n each connection returns to the side it started from; for odd n it ends on the other side.

edges $u_k u_{k+1}$ and the hooks of X_c and $X_{c'}$ at $u_1, \dots, u_{n-1}$, the ports at the two ends of the path are joined as follows:

$$\begin{aligned} &n \text{ even:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{L}_c(u_n), && \mathrm{R}_{c'}(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0); \\ &n \text{ odd:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{R}_{c'}(u_n), && \mathrm{L}_c(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0). \end{aligned}$$

These two connections use every port on the edges of the path, so no closed strand lies along the path. Moreover, for $0 < k < n$, the hook of X_c at u_k lies on the connection through $\mathrm{R}_c(u_0)$ if k is even. If k is odd, it lies on the connection through $\mathrm{L}_c(u_n)$ when n is odd, and on the connection through $\mathrm{R}_{c'}(u_n)$ when n is even.

Proof. Start at $\mathrm{R}_c(u_0)$, the port of X_c on u_0u_1 near u_0. The straight-through rule takes it to the port of $X_{c'}$ on u_0u_1 near u_1. In the orientation of $X_{c'}$, which runs from u_1 to u_0 along this edge, that port is $\mathrm{R}_{c'}(u_1)$. If $n = 1$ we have reached $\mathrm{R}_{c'}(u_n)$. Otherwise u_1 is a hook of $X_{c'}$, which leads to $\mathrm{L}_{c'}(u_1)$, the port of $X_{c'}$ on u_1u_2 near u_1. Crossing u_1u_2 leads to the port of X_c on u_1u_2 near u_2, which is $\mathrm{L}_c(u_2)$. If $n = 2$ we stop. Otherwise the hook of X_c at u_2 leads to $\mathrm{R}_c(u_2)$, and the pattern repeats with all indices shifted by two. So the path visits the hooks of $X_{c'}$ at odd k and the hooks of X_c at even k. It ends at $\mathrm{L}_c(u_n)$ if n is even and at $\mathrm{R}_{c'}(u_n)$ if n is odd. The same argument from the other end, with X_c and $X_{c'}$ exchanged (and, for odd n, L and R), starting at $\mathrm{R}_{c'}(u_n)$ for even n and at $\mathrm{L}_c(u_n)$ for odd n, gives the second connection; it visits the remaining hooks. Each connection crosses each of the n edges once, using two of its four ports, and the two connections use different ports, so together they use all of them. □

5.2. **Marks, runs and the skeleton.**

Definition 5.2 (Marks and runs)**.** A *mark* is a point that is a rim vertex of some child and is either

(i) a *junction*: a rim vertex of a child X_c at which the two rim edges of X_c have different neighbours, where a neighbour is a sibling or "outside P"; or
(ii) one of the crossover corners A_c, B_c of some child.

A *run* of X_c is a path along the rim of X_c between two consecutive marks lying on that rim. (The corners $A_c \neq B_c$ are marks on the rim of X_c, so runs are arcs with distinct ends.)

Along a run the neighbour of X_c does not change. So a run either lies on the rim of P (a *rim run*) or is shared with exactly one sibling $X_{c'}$ (an *interior run*). In the second case it is also a run of $X_{c'}$, its *partner*, traversed in the opposite direction, because marks are points and the same marks bound it on both sides. At an interior vertex u of an interior run, X_c and $X_{c'}$ share both edges at u, so the full angle at u is filled by these two children and no third child touches u. Since crossover corners are marks, the Zigzag Lemma applies to every interior run.

Definition 5.3 (Skeleton)**.** The *skeleton* $\mathcal{S}$ of the composition $P = \bigcup X_c$ consists of the following data: the children and their kinds; for each child, the cyclic sequence of its marks and runs, and which of its marks are A_c and B_c; for each interior run, its partner run and the parity of its length; the cyclic order of the rim runs round ∂P.

Two skeletons of compositions with the same children (by number and kind) are *equal* if there is a bijection of their marks that preserves, for each child, the cyclic order of its marks and the labels A_c, B_c, and preserves partner runs, the parities of interior runs and the cyclic order of the rim runs. The maps τ and σ and the parent ports below are defined from these data alone, so such a bijection carries them, and hence the conclusions of Proposition 5.4, from one composition to the other.

The *reduced ports* are the ports $\mathrm{R}_c(u)$ and $\mathrm{L}_c(u)$ for every child c and every mark u on its rim. The port $\mathrm{R}_c(u)$ lies on the run of X_c that starts at u, and $\mathrm{L}_c(u)$ on the run that ends at u. Define two involutions on reduced ports.

- *Inside a child*: $\tau(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$ if $u \notin \{A_c, B_c\}$; $\tau(\mathrm{R}_c(A_c)) = \mathrm{R}_c(B_c)$ and $\tau(\mathrm{L}_c(A_c)) = \mathrm{L}_c(B_c)$.
- *Across an interior run* from u to u' of X_c, of length n, with partner in $X_{c'}$: if n is even, $\sigma(\mathrm{R}_c(u)) = \mathrm{L}_c(u')$ and $\sigma(\mathrm{R}_{c'}(u')) = \mathrm{L}_{c'}(u)$; if n is odd, $\sigma(\mathrm{R}_c(u)) = \mathrm{R}_{c'}(u')$ and $\sigma(\mathrm{L}_c(u')) = \mathrm{L}_{c'}(u)$.

Both τ and σ are determined by $\mathcal{S}$. The *parent ports* are the reduced ports on rim runs. They are exactly the rim ports $\mathrm{R}(u), \mathrm{L}(u)$ of P at the rim vertices u of P that are marks (the *parent marks*), since X_c and P induce the same orientation on a rim run.

Proposition 5.4 (Reduced composition)**.** *Let $P = \bigcup X_c$, with every child having the signature. Then:*

(a) *every rim vertex of P that is not a mark is a hook of P;*
(b) *for every reduced port p, the ribbon of P followed from p (first through its child) reaches the rim of P at the first parent port among $\tau(p)$, $\tau\sigma\tau(p)$, $\tau\sigma\tau\sigma\tau(p), \ldots$, if there is one (the ports $\sigma\tau(p), \ldots$ in between lie on interior runs); for a parent port p this is $\pi_P(p)$;*
(c) *$\kappa(P)$ is the number of cycles of the alternation of σ and τ (none contains a parent port, since σ is not defined there).*

In particular, π_P at the parent marks and $\kappa(P)$ are determined by the skeleton.

Proof. (a) Let u be a rim vertex of P that is not a mark. The rim edge of P after u lies in some X_c; if the other rim edge of X_c at u were shared with a sibling, u would be a junction. So both rim edges of P at u belong to X_c, and u lies inside a rim run of X_c and is not a crossover corner of X_c. Hence $\pi_{X_c}(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$, and both ports are rim ports of P. (b) and (c) follow from Lemma 4.2: the ribbon starting at p runs through X_c to a reduced port $\tau(p)$, since by the signature of X_c the arc of X_c starting at a port at a mark ends at a port at a mark. If $\tau(p)$ lies on an interior run, the ribbon then follows that run, by the Zigzag Lemma, to $\sigma\tau(p)$. A closed strand of P not inside a child must cross some shared edge, and by the Zigzag Lemma it then passes through a reduced port; following it from there gives a cycle of the alternation. Conversely each such cycle, read as arcs of children joined by zigzag connections, is a closed strand of P, and distinct cycles give distinct strands. The children have no closed strands. □

Corollary 5.5 (Where a hook goes)**.** *In the situation of Proposition 5.4, let u be a rim vertex of X_c that is not a mark. If u lies inside a rim run, the hook of X_c at u is a hook of P. If u lies inside an interior run, traversed by X_c from u_0 to u_n, at distance k from u_0, then the hook of X_c at u lies on the same arc or closed strand of P as the reduced port $\mathrm{R}_c(u_0)$ if k is even; if k is odd, as $\mathrm{L}_c(u_n)$ when n is odd and as $\mathrm{R}_{c'}(u_n)$ when n is even.*

Proof. This is the last part of Lemma 5.1, together with Proposition 5.4(a). □

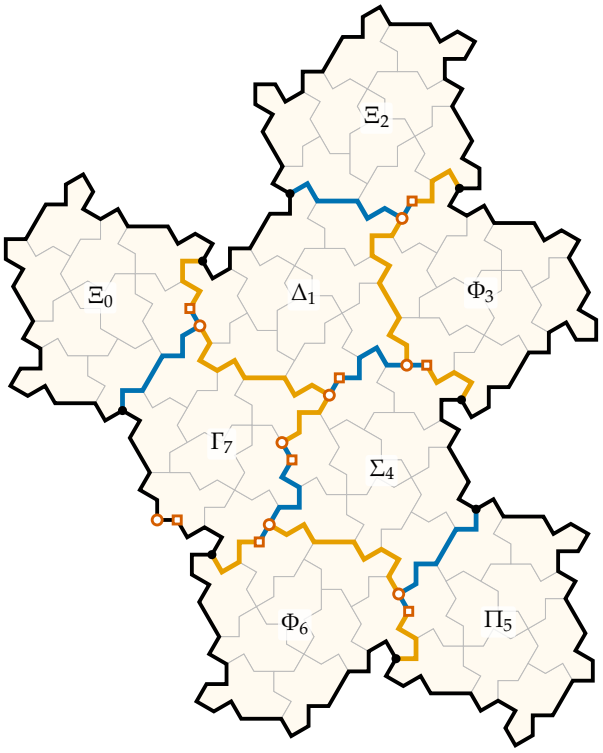


FIGURE 9. Marks and runs for the composition of a level-2 Δ-supertile from its eight level-1 children (subscripts are child numbers). Dots: the 24 marks (open circles and squares: the children's crossover corners A_c and B_c). Shared runs are coloured by the parity of their length: blue odd, orange even. Black: rim runs. The skeleton records this picture up to the lengths of the runs, keeping only the parities of the shared runs.

kind	side 0	side 1	side 2	side 3	side 4	side 5
Γ	α^-	α^+	γ^-	δ^-	β^+	β^-
Δ	γ^+	β^+	ϵ^-	α^+	γ^-	ζ^-
Θ	γ^+	β^+	θ^+	β^+	η	β^-
Λ	γ^+	β^+	ϵ^-	α^+	θ^-	β^-
Ξ	α^-	ϵ^+	θ^+	β^+	η	β^-
Π	α^-	ϵ^+	ϵ^-	α^+	θ^-	β^-
Σ	ζ^+	β^+	ϵ^-	α^+	γ^-	δ^+
Φ	γ^+	β^+	ϵ^-	ϵ^+	η	β^-
Ψ	α^-	ϵ^+	ϵ^-	ϵ^+	η	β^-

TABLE 4. The label of [19, Fig. 4.2] carried by side j of a supertile of each kind: the label of the edge of the hexagon in direction W, SW, SE, E, NE, NW for $j = 0, \dots, 5$.

6. SIDE LENGTHS

6.1. **Letters.** Every side of a level-m supertile ($m \geq 1$) carries one of the edge labels of [19, Fig. 4.2] (Table 4). Ignoring the signs, there are eight *letters*, $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$. Two sides that are glued to each other carry labels x^+ and x^- (or η and η), and so the same letter; this is the matching rule of [19, §4], and for sides glued inside a supertile it can be read off Tables 1, 2 and 4.

Proposition 6.1 (Lengths by letter)**.** *For every $m \geq 1$ the length $\ell_m(K, j)$ depends only on the letter of side j of K. Writing x_m for the common length of the sides with letter x,*

$$
\begin{aligned}
\alpha_{m+1} &= \beta_m + \epsilon_m + \eta_m, & \beta_{m+1} &= \alpha_m + \epsilon_m, \\
\gamma_{m+1} &= \beta_m + 2\epsilon_m + \eta_m + \theta_m, & \delta_{m+1} &= \alpha_m + \beta_m + \epsilon_m + 2\theta_m, \\
\epsilon_{m+1} &= 3\epsilon_m + \eta_m, & \zeta_{m+1} &= \alpha_m + \epsilon_m + \theta_m, \\
\eta_{m+1} &= 4\epsilon_m + \eta_m, & \theta_{m+1} &= 4\epsilon_m + \eta_m + \theta_m,
\end{aligned} \tag{1}
$$

with $(\alpha_1, \beta_1, \gamma_1, \delta_1, \epsilon_1, \zeta_1, \eta_1, \theta_1) = (7, 5, 10, 10, 8, 6, 10, 11)$.

Proof. For $m = 1$ this is a direct count on the level-1 supertiles (Fig. 3; Appendix A). By Proposition 2.4(c), $\ell_{m+1}(T, J)$ is the sum of the lengths of the children's sides listed in Table 3. Inspection of Tables 1, 3 and 4 shows that the multiset of letters of these children's sides depends only on the letter of side J of T, and is the one in (1). For example, side 0 of Δ (letter γ) is made of 7.5, 6.4, 6.3, 6.2, 5.4, with letters $\beta, \eta, \epsilon, \epsilon, \theta$. The claim follows by induction on m. □

The recurrence (1) has a surprisingly simple solution. Let F_n be the Fibonacci numbers, $F_1 = F_2 = 1$.

Proposition 6.2 (Fibonacci side lengths)**.** *For every $m \geq 1$,*

$$
\alpha_m = F_{3m+2} + 2, \qquad \beta_m = F_{3m+1} + 2, \qquad \gamma_m = \delta_m = \eta_m = 2F_{3m+2},
$$
$$
\epsilon_m = F_{3m+3}, \qquad \zeta_m = 2F_{3m+1}, \qquad \theta_m = F_{3m+4} - 2.
$$

Proof. For $m = 1$, $(F_4, F_5, F_6, F_7) = (3, 5, 8, 13)$ gives the initial values. For the induction step put $a = F_{3m+1}$ and $b = F_{3m+2}$, so that $F_{3m+3} = a + b$, $F_{3m+4} = a + 2b$, $F_{3m+5} = 2a + 3b$, $F_{3m+6} = 3a + 5b$ and $F_{3m+7} = 5a + 8b$. Substituting into (1):

$$
\alpha_{m+1} = (a+2) + (a+b) + 2b = F_{3m+5} + 2, \qquad \beta_{m+1} = (b+2) + (a+b) = F_{3m+4} + 2,
$$
$$
\gamma_{m+1} = (a+2) + 2(a+b) + 2b + (a+2b-2) = 4a + 6b = 2F_{3m+5},
$$
$$
\delta_{m+1} = (b+2) + (a+2) + (a+b) + 2(a+2b-2) = 4a + 6b,
$$
$$
\eta_{m+1} = 4(a+b) + 2b = 4a + 6b, \qquad \epsilon_{m+1} = 3(a+b) + 2b = F_{3m+6},
$$
$$
\zeta_{m+1} = (b+2) + (a+b) + (a+2b-2) = 2F_{3m+4},
$$
$$
\theta_{m+1} = 4(a+b) + 2b + (a+2b-2) = 5a + 8b - 2 = F_{3m+7} - 2. \qquad \square
$$

Since F_n is even exactly when $3 \mid n$:

Corollary 6.3 (Parities are constant)**.** *For every $m \geq 1$ the lengths α_m, β_m, θ_m are odd and γ_m, δ_m, ϵ_m, ζ_m, η_m are even.*

For example, the rim of a level-m Δ-supertile has length $E_m(\Delta) = \alpha_m + \beta_m + 2\gamma_m + \epsilon_m + \zeta_m$, which is $46, 182, 758, 3198, 13534, \ldots$.

6.2. **The crossover corner.**

Lemma 6.4 (Position of A)**.** *Let $m \geq 1$. For $K \neq \Sigma$ the crossover corner A of a level-m K-supertile is its vertex 0. For $K = \Sigma$ it lies on side 5 (letter δ), at distance*

$$
d_m = 2F_{3m+1}
$$

from vertex 5 and at distance $\delta_m - d_m = 2F_{3m}$ from vertex 0.

Proof. For $K \neq \Sigma$ this is Proposition 2.4(d) for $m \geq 2$, and is read off Fig. 3 for $m = 1$. For $m = 1$, $d_1 = 6$ is read off the level-1 Σ-supertile. By Proposition 2.4(d) and Table 3, the crossover corner of a level-$(m+1)$ Σ-supertile follows the children's sides 0.2, 0.1, 7.0 on its side 5. Child 0 is a Ξ and child 7 a Γ, so these sides have letters θ, ϵ, α and $d_{m+1} = \theta_m + \epsilon_m + \alpha_m = (a + 2b - 2) + (a + b) + (b + 2) = 2F_{3m+4}$, in the notation of the previous proof. □

Lemma 6.5 (Lower bounds)**.** *For every $m \geq 1$: every side has length at least 5; $d_m \geq 6$; and $\delta_m - d_m \geq 4$.*

Proof. By Proposition 6.2 all letter lengths increase with m, and at $m = 1$ the smallest is $\beta_1 = 5$. Also $d_m \geq d_1 = 6$ and $\delta_m - d_m = 2F_{3m} \geq 2F_3 = 4$. □

7. The signature at every level

This section proves the Signature Theorem and Theorem A. Section 7.1 describes how a supertile is glued from its children purely in terms of the side lengths ℓ_m and the offset d_m of Section 6. Section 7.2 shows, with the Zigzag Lemma, that lengthening a side by two edges preserves the signature of such a composition, and that the lengths at every level are reached from those at level 1 by such steps; so one finite computation at level 1 suffices. Section 7.3 runs the induction on the level, and Section 7.4 deduces Theorem A.

7.1. **Compositions as a function of side lengths.** Fix a parent kind T, with children $c = 0, \ldots, n_T - 1$ of kinds K_c (Table 1); exactly one of them, c_Σ, is a Σ. Let L assign a positive integer $L(c, j)$ to each child c and side $j \in \{0, \ldots, 5\}$, so that sides glued by Table 2 have the same length; let d be an integer with $2 \leq d \leq L(c_\Sigma, 5) - 2$; and let $\omega \in \{\pm 1\}$. The *composition* $C_T(L, d, \omega)$ is the following abstract patch of boxes. Child c has a rim of $E_c = \sum_j L(c, j)$ vertices, numbered $0, \ldots, E_c - 1$ in its own order; side j runs from vertex $V_c(j) = \sum_{i<j} L(c, i)$ to vertex $V_c(j + 1)$. Its crossover corner A_c is vertex 0, or for $c = c_\Sigma$ the vertex at distance d from vertex 5 along side 5, and $B_c = A_c + \omega$. Each child is a box with the signature at (A_c, B_c) (Definition 4.3). Sides are glued in pairs as in Table 2, reversing direction: the o-th edge of side j of c is the o-th edge from the end of side j' of c'. The sides marked ∂ form the rim of the composition, in the order of Table 3 read backwards (the table lists them in the parent's own order); a rim vertex of the composition is the point between two consecutive rim edges, and its ports R and L are the ports, on these two edges, of the children they belong to. Ribbons are followed as in Lemma 4.2: alternately through a child, by its signature, and straight across a glued edge, until the rim is reached. This defines the box of the composition: the pairing of its rim ports and its closed strands. Lemma 5.1, Proposition 5.4, and Corollary 5.5 and their proofs apply unchanged to compositions (with marks, runs and skeletons defined as in Definitions 5.2 and 5.3), since they use only the signatures of the children and the straight-through rule. The two facts that Section 5.2 reads off the angles at a vertex hold here by construction: the edges of glued sides are glued in reverse order, so a run of one child is a run of its partner; and the two sides of a child at one of its six vertices face different siblings, or the rim (Table 2), so no third child meets the inside of a run. We say that $C_T(L, d, \omega)$ *has the signature* if it has no closed strand, and every rim vertex is a hook except the crossover corners A_Γ, B_Γ of its Γ child, which are adjacent rim vertices (B_Γ following A_Γ in the children's order if $\omega = +1$, and preceding it if $\omega = -1$) and carry the crossover.

Write ℓ_m also for the function $(c, j) \mapsto \ell_m(K_c, j)$. By Propositions 2.4 and 6.1 and Lemma 6.4, if the children of a level-$(m + 1)$ T-supertile have the signature at $(A_c, A_c + \omega_m)$ in their own orientation, then the supertile is glued from its children as $C_T(\ell_m, d_m, \omega_m)$ is (with all children in their own orientation). Indeed: (i) by Proposition 2.4(a),(b) and Proposition 6.1 the children have the kinds and side lengths of the composition, and side j of X_c is side j' of $X_{c'}$ traversed oppositely, so points are matched as above; (ii) the rims of two children meet only in the closed sides they share: at a point inside a side both rim edges are shared with the partner, so the two children fill a neighbourhood, and a vertex lies on exactly three level-m supertiles, any two of which share a side ending there (Theorem 2.1(b)); (iii) so each side of a child is wholly shared with one sibling or wholly on the rim, as in Table 2; (iv) A_c is placed by Lemma 6.4, and $B_c = A_c + \omega_m$ by hypothesis; (v) the rim of the parent is given by Table 3 (Proposition 2.4(c)). Hence the children's rims, their corners and the gluing of their sides are those of $C_T(\ell_m, d_m, \omega_m)$. By

Lemma 4.2, the strands of the supertile, followed from child to child, are then those of the composition; in particular the box of the supertile is the box of $C_T(\ell_m, d_m, \omega_m)$.

7.2. **Lengthening.** Let (L, d) be as above. A *move* lengthens one side of one child by two edges, together with the side glued to it if there is one. An edge e of that side is replaced by a path of three edges through two new vertices; e must not be the first or the last edge of its side (in each child that contains it), nor one of the two edges at A_Σ (the crossover corner of the Σ child). The result is (L', d'), where L' is L with the side (and its partner) two edges longer, and $d' = d + 2$ if e lies on side 5 of the Σ child before A_Σ, and $d' = d$ otherwise.

Lemma 7.1 (Lengthening)**.** *If $C_T(L, d, \omega)$ has the signature, then so does $C_T(L', d', \omega)$, for every move $(L, d) \to (L', d')$.*

Proof. Write $C = C_T(L, d, \omega)$ and $C' = C_T(L', d', \omega)$, and let u_0u_1 be the edge e. In C' it is replaced by the path $u_0v_1v_2u_1$; positions after e on the rim of each child containing it move on by two, and every vertex of C has an image vertex in C'. The corners of C' are the images of those of C: for A_Σ this is the choice of d', and $B_c = A_c \pm 1$ stays next to A_c, because the edge between them (the first edge of side 0, the last of side 5, or an edge at A_Σ) is never cut. In particular v_1 and v_2 are not crossover corners. Let φ map each port of C to the port of C' of the same hand at the image vertex. Then φ commutes with the pairings inside the children, since the corners correspond, and with the straight-through rule across every glued edge other than e. If e is shared by children c and c', crossing it joins $\mathrm{R}_c(u_0)$ to $\mathrm{R}_{c'}(u_1)$ and $\mathrm{L}_c(u_1)$ to $\mathrm{L}_{c'}(u_0)$ (the Zigzag Lemma, Lemma 5.1, with $n = 1$); in C', by Lemma 5.1 with $n = 3$, whose interior vertices v_1, v_2 are hooks of both children, the ribbon from $\varphi(\mathrm{R}_c(u_0))$ runs along the new path to $\varphi(\mathrm{R}_{c'}(u_1))$, and the ribbon from $\varphi(\mathrm{L}_c(u_1))$ to $\varphi(\mathrm{L}_{c'}(u_0))$. If e lies on the rim, v_1 and v_2 are hooks of the child, and so new rim vertices of C' that are hooks of C'.

Hence, for every port x of C, the ribbon of C' from $\varphi(x)$ passes through the images of the ports met from x in C, and reaches the rim at the image of the rim port reached in C. So the pairing of the rim ports of C' at the images of the rim vertices of C is the image of that of C: a hook at every one of them except A_Γ and B_Γ, which are adjacent as before and carry the crossover; and the new rim vertices, if any, are hooks. Finally, C' has no closed strand. Every port of C' at an old vertex is some $\varphi(x)$, whose ribbon reaches the rim. A strand through a port at v_1 or v_2 either is a hook of the rim (if e lies on the rim), or follows one of the two connections of Lemma 5.1 along $u_0v_1v_2u_1$, which use every port on these edges and end at ports at u_0 and u_1; so it too reaches the rim. □

Call (L, d) *admissible* if sides glued by Table 2 have the same length; $L(c, j) \geq \ell_1(c, j)$ and $L(c, j) \equiv \ell_1(c, j) \pmod 2$ for all c and j; $d \geq d_1$ and $d \equiv d_1 \pmod 2$; and $L(c_\Sigma, 5) - d \geq \ell_1(c_\Sigma, 5) - d_1 = 4$. Then $d \geq 6$ and $L(c_\Sigma, 5) - d \geq 4$, so the composition $C_T(L, d, \omega)$ is defined.

Lemma 7.2 (Reachability)**.** (a) *For every $m \geq 1$, the pair (ℓ_m, d_m) is admissible.*

(b) *Every admissible $(L, d) \neq (\ell_1, d_1)$ arises by a move from an admissible pair (L_0, d_0) with $\sum_{c,j} L_0(c, j) < \sum_{c,j} L(c, j)$. The move can be chosen so that its edge e lies at least two edges from both ends of its side, in each child that contains it, unless that side is side 5 of the Σ child or is glued to it.*

Hence every admissible pair is reached from (ℓ_1, d_1) by finitely many such moves through admissible pairs.

Proof. (a) Glued sides have the same letter (Section 6), so the same length. By Proposition 6.2 the letter lengths do not decrease with m, and by Corollary 6.3 their parities do not change. By Lemma 6.4, $d_m = 2F_{3m+1} \geq 6 = d_1$ is even, and $\ell_m(c_\Sigma, 5) - d_m = \delta_m - d_m = 2F_{3m} \geq 4$.

(b) If $L = \ell_1$, then $L(c_\Sigma, 5) = \delta_1 = 10$ and $d_1 \leq d \leq 10 - 4$, so $d = d_1$. Hence $L \neq \ell_1$, and by the parity condition some side j of some child c has $L(c, j) \geq \ell_1(c, j) + 2$. We undo a move on that side. If the side is neither side 5 of Σ nor glued to it, take $d_0 = d$ and the move whose edge is the third edge of the side (the third from the end in the partner); before the move

the side has length at least $\ell_1(c, j) \geq 5$ (Lemma 6.5), so this edge lies at least two edges from both ends, in either child. Otherwise we undo a move on side 5 of the Σ child (and the side glued to it, which belongs to another child), measuring positions along side 5 from its vertex 5. If $d \geq d_1 + 2$, take $d_0 = d - 2$ and the move whose edge is the second edge of the side, before A_Σ; it does not touch A_Σ, since $d_0 \geq 6$. If $d = d_1$, take $d_0 = d$ and the second edge after A_Σ: before the move the side has $L(c_\Sigma, 5) - 2 - d \geq \ell_1(c_\Sigma, 5) - d_1 = 4$ edges after A_Σ, so this edge is at least two edges from the end of the side. In each case (L_0, d_0) is admissible, and $\sum L_0 = \sum L - 2$ or $\sum L - 4$. □

Computation 7.3 (The compositions at the level-1 lengths)**.** For each of the nine kinds T and both signs ω, the composition $C_T(\ell_1, d_1, \omega)$ has the signature.

This was computed twice: by a program that builds the 18 compositions from Tables 1 to 3 and the lengths alone, follows the ribbon from every one of their ports and checks each condition of the signature (Appendix A), and in the kernel of the Lean proof assistant (Section 1, *Formal verification*). Each composition has 8 children (7 for Γ), 732 ports (640) and a rim of 182 edges (160). As a negative control, with each child's corner B_c placed two vertices from A_c instead of one, all 18 checks fail, in both programs.

Proposition 7.4 (Compositions at every level)**.** *For every $m \geq 1$, every kind T and both signs ω, the composition $C_T(\ell_m, d_m, \omega)$ has the signature.*

Proof. By Computation 7.3 and Lemma 7.1, every composition reached from (ℓ_1, d_1) by moves has the signature. By Lemma 7.2 this includes $C_T(\ell_m, d_m, \omega)$. □

7.3. **The induction.**

Computation 7.5 (Base case)**.** Every level-1 supertile has the signature at (A, B), where A is as in Lemma 6.4 and $B = A + \omega_1$ in own order. Moreover ℓ_1 and d_1 are as stated in Proposition 6.1 and Lemma 6.4.

This was computed from the tiles, using Lemma 4.2, in exact arithmetic; the program checks the statement as worded: the two non-hooks are the crossover corners of the Γ child (the Mystic), they are adjacent, $B = A + \omega_1$, they carry the crossover, and there is no closed strand. A second, separate program, in floating point, confirms it (Appendix A). The level-1 supertiles, as glued patches of tiles, are also the input of the Lean formalisation, whose kernel checks the statement again, with A placed as in Lemma 6.4 (the formalisation does not single out the Mystic). The rims have $E_1 = 44$ edges for Γ and 46 for the other kinds.

Theorem 7.6 (Signature Theorem)**.** *In every homochiral tiling by* Tile(1, 1)*, every level-m supertile of every kind has the signature at (A, B), where A is its crossover corner (Section 2.3; placed as in Lemma 6.4 for $m \geq 1$) and $B = A + \omega_m$ in its own order. In particular no supertile contains a closed strand.*

Proof. For $m = 0$ this is Lemma 4.4, and for $m = 1$ it is Computation 7.5. Let $m \geq 1$, assume the theorem for level m, and let Y be a level-$(m+1)$ supertile of kind T. Its children have the signature at $(A_c, A_c + \omega_m)$, so by Section 7.1 its box is that of $C_T(\ell_m, d_m, \omega_m)$, which has the signature by Proposition 7.4. Hence Y has no closed strand, and hooks at all its rim vertices except the crossover corners A_Γ, B_Γ of its Γ child, which are adjacent and carry the crossover. By definition A_Γ is the crossover corner of Y, placed as in Lemma 6.4. Finally $B_\Gamma = A_\Gamma + \omega_m$ in the child's own order, which is opposite to that of Y; so $B = A - \omega_m = A + \omega_{m+1}$ in the own order of Y. □

Remark 7.7. Since the own orientation alternates with m and so does ω_m, in absolute terms B is always the counter-clockwise neighbour of A on the rim, as it is for the tile ($B = v_1$).

Remark 7.8 (Other checks of the signature). Of the computations in this section the proof uses only Computations 7.3 and 7.5. The following computations and arguments agree

with them, and are not used. (1) The compositions $C_T(\ell_m, d_m, \pm 1)$ were built and followed port by port for $m \le 4$; each of the 1,792 single moves from (ℓ_1, d_1) was checked to keep the signature, with both signs ω; and the moves that the proof of Lemma 7.2 chooses were followed from (ℓ_2, d_2) and (ℓ_3, d_3) back to (ℓ_1, d_1) for every kind, each one checked to be a move and, from (ℓ_2, d_2), every composition on the way checked to have the signature. (2) The signature of the actual supertiles was computed from the tiles at levels 1 to 3 (and 4) in exact arithmetic, and at levels 1 to 9 in floating point, by the two programs of Appendix A. (3) A second route to Proposition 7.4 uses the skeleton (Definition 5.3). For letter lengths λ (and $C_T(\lambda, d, \omega)$ the composition with $L(c, j) = \lambda$ of the letter of side j of K_c) with $\lambda(x) \ge 3$ and $3 \le d \le \lambda(\delta) - 3$, the marks on each side of a child lie at the symbolic positions $0 < 1 < d-1 < d < d+1 < \ell-1 < \ell$ or $0 < 1 < \ell-d-1 < \ell-d < \ell-d+1 < \ell-1 < \ell$, whose order does not depend on the lengths, and the length of a run is a sum of side lengths plus a difference of symbolic positions. So the skeleton of $C_T(\lambda, d, \omega)$ depends only on λ mod 2, d mod 2 and ω, and by Proposition 5.4 so do its pairing at the parent marks and its closed strands (every other parent rim vertex is a hook). The 18 skeletons at $(\ell_1, d_1, \pm 1)$ have 24 marks (21 for Γ) and 52 runs (45), of which 10 (9) lie on the rim; their reduced walks give the signature, and they coincide with the skeletons of the actual supertiles at every level from 2 to 9 (a comparison that does not distinguish the parent's A from its B; the children's are distinguished).

7.4. **No closed strands.**

Proof of Theorem A. Let $\mathcal{T}$ be a homochiral tiling whose hierarchy covers the plane: for some tile t, the supertiles $X_0(t) \subset X_1(t) \subset \cdots$ containing t have union $\mathbb{R}^2$. Every tile t' lies in some $X_m(t)$: a point x of the interior of t' lies in $X_m(t)$ for some m, hence in some tile s of $X_m(t)$. Every neighbourhood of x meets the interior of s, so the interiors of s and t' meet, and $s = t'$. A closed strand consists of finitely many pieces in finitely many tiles, so it lies in a single $X_m(t)$, and all its joins are across interior edges of $X_m(t)$. It is then a closed strand of $X_m(t)$, contradicting Theorem 7.6. (Equally, each edge of a tile is shared with exactly one other tile, and the two lie in some $X_m(t)$, inside which the edge is interior. This is the form the formalisation uses.) By Lemma 3.4, every strand is a two-way infinite path. □

8. The nest tiling

This section builds the nest tiling $\mathcal{N}$, grown from one tile by taking each block as child 1 of the next, and shows that it is a tiling of the plane with no closed strand (Proposition 8.4). Child 1 of every supertile is a Δ (Table 1).

Construction 8.1 (The nest)**.** Let $\mathcal{T}$ be a homochiral tiling [19, Thm. 2.2]. For each $h \ge 0$, let $Y^{(h)}$ be a level-h Δ-supertile of $\mathcal{T}$; one exists, for instance child 1 of any level-$(h+1)$ supertile. Let $Y^{(h)} = Y^{(h)}_h \supset Y^{(h)}_{h-1} \supset \cdots \supset Y^{(h)}_0$ be its chain of descendants, each child 1 of the next. Let g_h be the proper motion taking the tile $Y^{(h)}_0$ onto Tile(1, 1) (with v_0 at the origin), and put $N^{(h)}_b = g_h(Y^{(h)}_b)$ for $b \le h$.

Proposition 8.2. *$N^{(h)}_b$ does not depend on $h \ge b$ (or on the choices made). Writing N_b for it, $N_0 = \{\text{Tile}(1,1)\}$, and N_b is child 1 of N_{b+1} for every b.*

Proof. Let $h, h' \ge b$, with $Y^{(h)}$, $Y^{(h')}$, g_h, $g_{h'}$ as in Construction 8.1. By Lemma 2.6 there is a proper motion f with $f(Y^{(h)}_b) = Y^{(h')}_b$ that maps descendants to descendants with their numbers; hence $f(Y^{(h)}_{b'}) = Y^{(h')}_{b'}$ for all $b' \le b$, in particular for the tiles Y_0. So $g_{h'} f$ and g_h both take the tile $Y^{(h)}_0$ onto Tile(1, 1), and since a tile has no symmetry they are equal. Thus $N^{(h')}_b = g_{h'} f(Y^{(h)}_b) = N^{(h)}_b$. □

The *nest tiling* is $\mathcal{N} = \bigcup_{b \ge 0} N_b$, a set of tiles. Figure 10 shows three consecutive blocks.

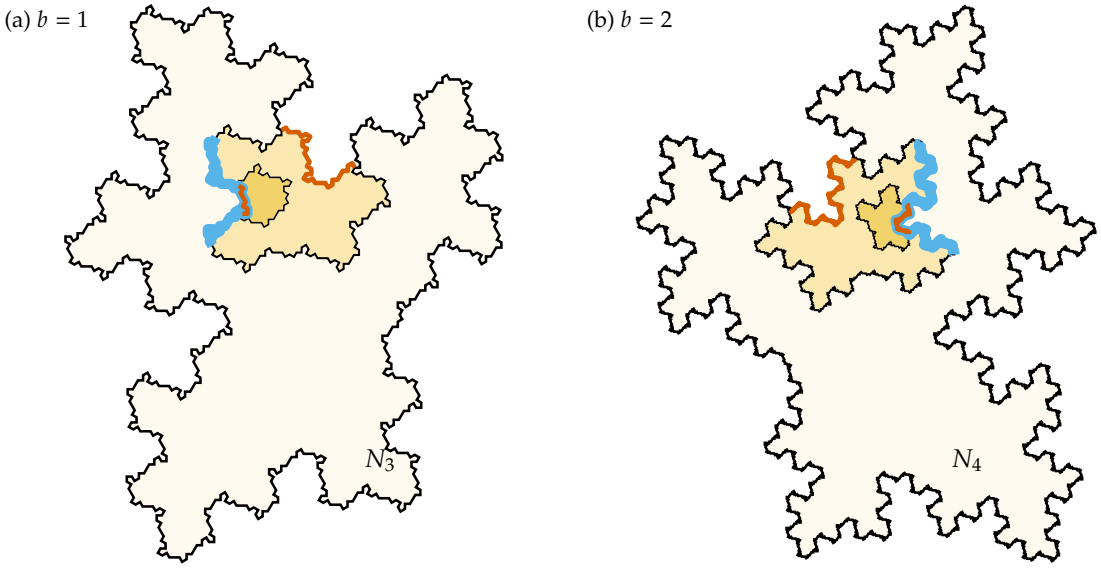


FIGURE 10. Three consecutive blocks of the nest: (a) $N_1 \subset N_2 \subset N_3$ and (b) $N_2 \subset N_3 \subset N_4$. Vermilion: side 2 of the two smaller blocks, which is the only side of each block on the rim of the next. Blue: side 4 of the middle block. Since the vermilion side of the smallest block lies inside the blue side, and the blue side faces a sibling inside the largest block, the smallest block does not reach the rim of the largest (Lemma 8.3).

Lemma 8.3 (Grandparent Lemma)**.** *Let $b \geq 1$.* (a) *The block N_b does not meet the rim of N_{b+2}.* (b) *Every edge of a tile of N_b is shared by two tiles of N_{b+2}.*

Proof. (a) N_b is child 1 of N_{b+1} (Proposition 8.2). By Table 2, the sides $0, 1, 3, 4, 5$ of child 1 face the siblings $7, 0, 2, 3, 4$, and side 2 lies on the parent's rim. A point of N_b that is not on its closed side 2 is therefore an interior point of N_{b+1}. This is clear for interior points of N_b and for interior points of the other sides, where N_b and one sibling together fill a neighbourhood. At the vertices $0, 1, 4, 5$ of N_b, exactly three level-b supertiles meet; two of them are the distinct siblings across the adjacent sides, and together they fill a neighbourhood. Hence

$$N_b \cap \partial N_{b+1} = \text{closed side 2 of } N_b. \tag{2}$$

By Table 3 (the Δ row: positions 14 to 18), side 4 of the parent is the concatenation of $2.2, 2.1, 1.2, 0.4, 0.3$. Since 1.2 is neither first nor last, the closed side 2 of N_b lies in the relative interior of side 4 of N_{b+1}. Now let $p \in N_b \cap \partial N_{b+2}$. Then $p \in \partial N_{b+1}$, for otherwise p would be interior to $N_{b+1} \subset N_{b+2}$. So p lies on side 4 of N_{b+1} by the above, and, by (2) for N_{b+1}, on the closed side 2 of N_{b+1}. But the closed sides 2 and 4 of N_{b+1} are disjoint: they are arcs of a simple closed curve separated by side 3 on one side and by sides $5, 0, 1$ on the other.

(b) follows from (a). It also follows from the tables alone, and this is how the formalisation proves it: a rim edge of N_b lies on a side $j \neq 2$ of N_b, which faces a sibling inside N_{b+1}, or on side 2, which lies on side 4 of N_{b+1} (position 16 of Table 3); and side 4 of child 1 faces sibling 3 inside N_{b+2} (Table 2). □

Proposition 8.4. *$\mathcal{N}$ is a homochiral tiling of the plane, and every tile of $\mathcal{N}$ lies in some N_b.*

Proof. Any two tiles of $\mathcal{N}$ lie in a common block, so their interiors are disjoint. We show that the tiles cover the plane. Fix h and work in the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$ of Construction 8.1, which contains $N_0, \ldots, N_h$ (Proposition 8.2). Call two tiles of $\mathcal{T}_h$ *neighbours*

b	tiles in N_b	arcs of N_b	$k=1$	$k=2$	$k=3$	$k=4$
0	1	14	5	2	1	1
1	9	46	10	2	2	1
2	71	182	37	2	1	1
3	559	758	146	2	2	1
4	4,401	3,198	613	2	1	1
5	34,649	13,534	2,586	2	2	1
6	272,791	57,318	10,949	2	1	–
even $b \geq 2$				2	1	1
odd $b \geq 1$				2	2	1

TABLE 5. The number of arcs of N_{b+k} on which the arcs of N_b lie, that is, the number of blocks of $\mathcal{P}_k(b)$ (row $b = 0$: the 14 pieces of the single tile, computed on the drawing up to level 5). Rows $1 \leq b \leq 6$: computed by following every ribbon, up to level 9 (in floating point; for $b \leq 3$ also in exact arithmetic, up to level 7). Last two rows: the proof of Theorem 9.1, for every $b \geq 1$ of each parity.

if they share a point, and let dist be the resulting graph distance. We claim that every tile at distance at most k from N_1 lies in N_{1+2k}, provided $1 + 2k \leq h$. For $k = 0$ there is nothing to prove. If t has distance $k + 1$, it shares a point p with a tile of N_{1+2k}. By Lemma 8.3, p is an interior point of N_{3+2k}, so the interior of t meets the interior of a tile of N_{3+2k}; hence t is that tile. Now let $x \in \mathbb{R}^2$ and let D be the disc of radius $r = |x| + 1$ about the origin. The tiles of $\mathcal{T}_h$ meeting D have diameter $3 + \sqrt{3} < 5$ and area $3 + 3\sqrt{3} > 8$, so there are at most $k_0 = \lfloor \pi(r + 5)^2/8 \rfloor$ of them. They cover the connected set D, so their neighbour graph is connected. One of them is $N_0 \subset N_1$, since the origin is a vertex of N_0. So every tile of $\mathcal{T}_h$ meeting D, in particular one containing x, lies in N_{1+2k_0} as soon as $h \geq 1 + 2k_0$. Since N_{1+2k_0} does not depend on h, x is covered by $\mathcal{N}$. □

The blocks N_b are supertiles in the sense of Section 2: each is congruent, with all its structure, to a level-b Δ-supertile of $\mathcal{T}$. So Theorem 7.6 applies to them. In particular $\mathcal{N}$ has no closed strand: a closed strand passes through finitely many tiles, which lie in a common block N_b by Proposition 8.4, and would be a closed strand of N_b.

9. MERGING AND THE SINGLE STRAND

This section shows that the arcs of every block of the nest merge within three or four levels (Section 9.1), and proves Theorem B (Section 9.2). The merging uses the device of Section 7: a map computed once at the level-1 lengths and carried to every level by the moves of Lemma 7.2.

9.1. **Merging.** For $b \geq 1$ and $k \geq 0$, every arc of N_b is contained in a unique arc of N_{b+k}, because N_{b+k} has no closed strand. Let $\mathcal{P}_k(b)$ be the partition of the $E_b(\Delta)$ arcs of N_b in which two arcs are in the same block when they lie on the same arc of N_{b+k}. The blocks only merge as k increases.

Theorem 9.1 (Merging)**.** *For every $b \geq 1$, $\mathcal{P}_4(b)$ has a single block; if b is even, so does $\mathcal{P}_3(b)$.*

Table 5 shows the numbers. The proof looks one level up at a time. In N_{m+1}, every arc of its child N_m lies on one of a few hooks of N_{m+1}, and which one depends only on where the arc leaves N_m: for most of the rim, only on the side of N_m. We compute this map once, at the level-1 lengths, and carry it to every level by the moves of Section 7.

arc of N_m	m even ($\omega_m = +1$)	m odd ($\omega_m = -1$)
crossover arc through R(A) and R(B)	17	21
crossover arc through L(A) and L(B)	9	16
hook at u, $0 < u \le V_1$	17	21
hook at u, $V_1 < u \le V_2$	17	17
hook at u, $V_2 < u < V_3$	a hook of N_{m+1} itself	
hook at u, $V_3 \le u < V_4$	16	16
hook at u, $V_4 \le u < V_5$	12	16
hook at u, $V_5 \le u < E$	9	16

TABLE 6. The junction map (Lemma 9.3): the junction q such that an arc of N_m lies on the hook H_q of N_{m+1}. Hooks are listed by the position u of their rim vertex, which is never A or B.

Junctions. Let $m \ge 1$. The rim of N_{m+1}, in its own order from its corner A, is the concatenation of the 22 children's sides listed in Table 3 for eight children, at positions $q = 0, \dots, 21$, with its sides starting where the Δ row says. The *junction* q is the rim vertex of N_{m+1} at which the side at position q starts (not to be confused with the junctions of Definition 5.2(i)). The junctions $9, 12, 16, 17, 21$ lie inside sides of N_{m+1} (Lemma 9.4 below), so they are hooks of N_{m+1} (Theorem 7.6); we write H_q for the hook arc of N_{m+1} at junction q. The side at position 16 is 1.2, side 2 of N_m, so junctions 16 and 17 are its two ends: vertices 3 and 2 of N_m. Positions on the rim of N_m are taken in its own order from its corner A, so that its vertices $0, \dots, 5$ lie at positions $0 = V_0 < V_1 < \cdots < V_5 < E$, where $E = E_m(\Delta)$; its corner $B = A + \omega_m$ is at position 1 if m is even and $E - 1$ if m is odd.

Computation 9.2 (The junction map at the level-1 lengths)**.** In $C_\Delta(\ell_1, d_1, \omega)$, for both signs ω, with child 1 in the place of N_m and positions and junctions defined in the same way: every rim vertex of child 1 strictly inside its side 2 is a hook of the composition; and the ribbon from every other port of child 1 reaches the rim of the composition at junction q, where q is given by Table 6 with $\omega_m = \omega$.

This was computed twice, like Computation 7.3: by the program of Appendix A, and in the kernel of the Lean proof assistant (its second part; the formalisation proves the first for every level, at the level of tiles). As a negative control, exchanging the junctions 12 and 16 in Table 6 makes the check fail for both signs, in both programs.

Lemma 9.3 (The junction map)**.** *Let $m \ge 1$. In N_{m+1}, every hook of N_m at a rim vertex strictly inside side 2 of N_m is a hook of N_{m+1}, and every other arc of N_m lies on the hook H_q of N_{m+1}, where q is given by Table 6.*

Proof. Say that (L, d) has the *junction property* if the statement of Computation 9.2 holds for $C_\Delta(L, d, \omega)$ for both ω. The children of N_{m+1} have the signature (Theorem 7.6), so by Section 7.1 N_{m+1} is glued from them as $C_\Delta(\ell_m, d_m, \omega_m)$ is, with N_m as child 1; and a ribbon of the composition from a port p of child 1 runs through child 1 along the arc of N_m ending at p, and on along the arc of N_{m+1} that contains it. A hook arc has both its ends at the same rim vertex. So the lemma says that (ℓ_m, d_m) has the junction property. By Computation 9.2, (ℓ_1, d_1) has it. By Lemma 7.2, (ℓ_m, d_m) is reached from (ℓ_1, d_1) by moves as in Lemma 7.2(b) through admissible pairs, so it suffices to show that such a move keeps the junction property. Child 1 is a Δ, and the side glued to side 5 of the Σ child (child 4) belongs to child 7 (Table 2); so a move whose edge lies on the rim of child 1 keeps it at least two edges from both ends of its side.

Let C, C' and φ be as in the proof of Lemma 7.1. Both have the signature (by the proof of Proposition 7.4), and the junctions of C' are the images of those of C, since junctions are vertices of children; so are the vertices V_j of child 1, and φ keeps the order of positions on its rim. For a port x of child 1 at an old vertex, the ribbon from $\varphi(x)$ reaches the rim at

the image of the rim port reached from x, and $\varphi(x)$ lies in the same row of Table 6 as x. It remains to consider the new vertices v_1, v_2 when the edge $e = u_0u_1$ lies on the rim of child 1. Then v_1, v_2, u_0, u_1 lie strictly inside the same side of child 1, at least two edges from its ends, so in the same row of Table 6, and none is a crossover corner. If e lies on the rim of the composition (that is, on side 2 of child 1), v_1 and v_2 are hooks of C', as the table says. Otherwise, by the last part of the Zigzag Lemma (Lemma 5.1, with $n = 3$), the hooks of child 1 at v_1 and v_2 lie on the strands of C' through φ of the ports of child 1 at u_1 and u_0. Those ports lie in the same row, so these strands are arcs of C' with an end at the junction that the row names; an arc with an end at a hook is that hook, and both its ends are at the junction (junction q is a hook of C': C' has the signature, and junction q is at least one child side, so at least five edges, along the rim from junction 0, which is A_Γ, so it is neither A_Γ nor B_Γ). So the ribbons from the ports at v_1 and v_2 reach the rim there too. □

Lemma 9.4 (Where the junctions lie)**.** *Let $m \geq 1$. The junctions* 9, 12, 16, 17 *and* 21 *of N_{m+1} lie on its sides* 2, 3, 4, 4 *and* 5 *respectively, at least five edges from both ends of the side. Every rim vertex strictly inside side* 2 *of N_m lies on side* 4 *of N_{m+1}, at least five edges from both ends.*

Proof. By Table 3, side 2 of a Δ is made of the children's sides at positions 7 to 10, side 3 of those at 11 to 13, side 4 of those at 14 to 18, and side 5 of those at 19 to 21. Junction q lies between the sides at positions $q - 1$ and q, which lie in the same side of N_{m+1} for $q = 9, 12, 16, 17, 21$; and side 2 of N_m is the side at position 16. Every side of a supertile of level ≥ 1 has at least five edges (Lemma 6.5). □

Proof of Theorem 9.1. A rim vertex of N_m at least two edges from both ends of its side J is not a crossover corner, and by Lemma 9.3 its hook lies on the hook of N_{m+1} that Table 6 names for side J: for even m, junction 17, 17, 16, 12, 9 for $J = 0, 1, 3, 4, 5$; for odd m, junction 21, 17, 16, 16, 16; and for $J = 2$, the vertex is itself a hook of N_{m+1}, on its side 4. By Lemma 9.4, each of these hooks of N_{m+1} lies at least five edges inside a side of N_{m+1}, so the same rule applies to it one level up.

Let $b \geq 1$ be even, so that $\omega_b = +1$. By Lemma 9.3 every arc of N_b lies on H_{17}, H_{16}, H_{12} or H_9 of N_{b+1}, or on a hook of N_{b+1} strictly inside side 2 of N_b; by Lemma 9.4 these are hooks inside sides 4, 4, 3, 2 and 4 of N_{b+1}. Since $b + 1$ is odd, the hooks inside sides 3 and 4 lie on H_{16} of N_{b+2}, and those inside side 2 are hooks of N_{b+2} inside its side 4. All are inside side 4 of N_{b+2}, and since $b + 2$ is even they lie on H_{12} of N_{b+3}. So $\mathcal{P}_3(b)$ has a single block.

Let $b \geq 1$ be odd. Every arc of N_b lies on H_{21}, H_{17} or H_{16} of N_{b+1}, or on a hook of N_{b+1} strictly inside side 2 of N_b: inside sides 5, 4, 4, 4 of N_{b+1}. Since $b + 1$ is even, those inside side 4 lie on H_{12} and those inside side 5 on H_9 of N_{b+2}, inside its sides 3 and 2. Since $b + 2$ is odd, the first lie on H_{16} of N_{b+3}, and the second are hooks of N_{b+3} inside its side 4. All are inside side 4 of N_{b+3}, and since $b + 3$ is even they lie on H_{12} of N_{b+4}. So $\mathcal{P}_4(b)$ has a single block. Once $\mathcal{P}_k(b)$ has a single block, so does $\mathcal{P}_{k'}(b)$ for every $k' \geq k$.

The same steps give the last two rows of Table 5. For even b the arcs of N_b lie on two arcs of N_{b+2}: H_{16}, and the hook at junction 9 of N_{b+1}, which lies inside side 2 of N_{b+1} and so is not junction 16 of N_{b+2} (an end of that side). For odd b they lie on H_{12} and H_9 of N_{b+2}, and on H_{16} of N_{b+3} and the hook at junction 9 of N_{b+2}, which for the same reason is not junction 16 of N_{b+3}. Both arcs are reached in each case (for instance from vertices well inside sides 4 and 5 of N_b, or sides 4 and 0), and distinct hooks are distinct arcs. □

Remark 9.5 (Other checks of Theorem 9.1). Besides Computations 7.3 and 7.5, the proof of Theorem 9.1 uses only Computation 9.2. The following agree with it, and are not used. (1) Table 5 was computed on the actual blocks for $b \leq 6$, following every ribbon up to level 9, and for $b \leq 3$ by a second program in exact arithmetic up to level 7. (2) The junction map was checked directly on $C_\Delta(\ell_m, d_m, \pm 1)$ for $m \leq 4$, after each of the 190 single moves from (ℓ_1, d_1) on the Δ composition whose edge, if it lies on child 1, is at least two edges from the ends of its side (with both signs), and on the actual blocks $N_1 \subset N_2$, $N_2 \subset N_3$, $N_3 \subset N_4$, built from the tiles in exact arithmetic. (3) A second proof follows the arcs of N_b up the nest

with *label states*: finite records of which hooks and crossover arcs of N_{b+k} carry arcs of N_b, advanced one level at a time with the reduced walk (Proposition 5.4) and Corollary 5.5, with every position written as an integer affine function of ℓ_{b-1}. Run symbolically over 8 relative levels, once for each parity of b, its branch decisions are 2,106 linear conditions on the side lengths (1,407 distinct). For every $b \geq 2$, 1,980 of them hold because every side has at least 5 edges, and the other 126 were certified twice: from Proposition 6.2 with Binet's formula, and by exact rational certificates on the space spanned by the side-length vectors ℓ_1, ℓ_2, ℓ_3, which contains every ℓ_{b-1}. With $b = 1$ computed directly, this gives Theorem 9.1 and the last two rows of Table 5 again.

9.2. **One strand.**

Proof of Theorem B. By Lemma 3.4, every component of the strand graph $\mathcal{G}(\mathcal{N})$ is a cycle or a two-way infinite path, and there are no cycles, as shown after Proposition 8.4. It remains to show that $\mathcal{G}(\mathcal{N})$ is connected. Let p, q be pieces of tiles t, t' of $\mathcal{N}$. By Proposition 8.4 both tiles lie in some N_b with $b \geq 1$. Since N_b has no closed strand, p and q lie on arcs of N_b. By Theorem 9.1 these arcs lie on one arc of N_{b+4}, so p and q are on the same strand. The strand passes through every tile, since every tile has pieces. □

10. Scope and open questions

Other tilings. Theorem A holds for every tiling whose hierarchy covers the plane. Theorem B is proved for the nest tiling, whose chain of supertiles is $\Delta \subset \Delta \subset \cdots$ with child number 1 at every step. Every tiling whose hierarchy covers the plane has some such chain $(K_m, c_m)_{m \geq 1}$, where X_m is a level-m supertile of kind K_m and child c_m of X_{m+1}. The induction of Section 7 does not see the chain, but the merging of Section 9 does: its finite part depends on the sequence (K_m, c_m), and so may its outcome.

Remark 10.1 (Three strands: a computation, not a theorem). Let $K_m = \Theta$ for odd m and $K_m = \Gamma$ for even m, with $c_m = 2$ for odd m and $c_m = 7$ for even m (Table 1). A tiling with this chain is constructed as in Construction 8.1, and its hierarchy covers the plane: the argument of Lemma 8.3 and Proposition 8.4, applied with Tables 2 and 3, shows that no X_m meets the rim of X_{m+2}. In this tiling, following the ribbons through the actual supertiles in exact arithmetic, the arcs of X_1 lie on exactly three arcs of X_m for $3 \leq m \leq 8$, and the same holds for the arcs of X_2 and of X_3 for $5 \leq m \leq 8$. The label bookkeeping of Remark 9.5, run along this chain, agrees and still gives three at $m = 20$. If this persists for every block and every level, the pieces of this tiling form exactly three strands, each a two-way infinite path by Theorem A. We have not proved this, and the paper does not use it. For the other alignment ($K_m = \Gamma$, $c_m = 7$ for odd m and $K_m = \Theta$, $c_m = 2$ for even m) the arcs of X_1 lie on a single arc of X_4. Along 90,000 chains chosen at random step by step (30,000 for each $b \leq 3$), the label bookkeeping put the arcs of X_b on a single arc of X_m for some $m \leq 20$ in all but five. The programs are in `experiments/`, with logs; the label bookkeeping along a chain is `hierarchy/chains_label.js`, which agrees with exact strand following on 60 chains up to level 8.

(1) Which homochiral tilings whose hierarchy covers the plane carry a single strand? Is the number of strands always finite, and which numbers occur?

Faults. Theorem A assumes that the hierarchy covers the plane. This subsection says what that leaves out, and why.

Take a tile t and the supertiles $X_0(t) \subset X_1(t) \subset \cdots$ that contain it. If the hierarchy covers the plane, they grow until they cover the whole plane. Then every closed strand, being made of finitely many pieces, lies inside one of them, and the Signature Theorem (Theorem 7.6) rules it out: that is the whole proof of Theorem A (Section 7.4). In a tiling with a *fault* they grow for ever but fill only part of the plane, an *infinite supertile* (Tatham's term [20, §5]; Goodman-Strauss [10] speaks of infinitely large supertiles). The rest of the plane is filled by

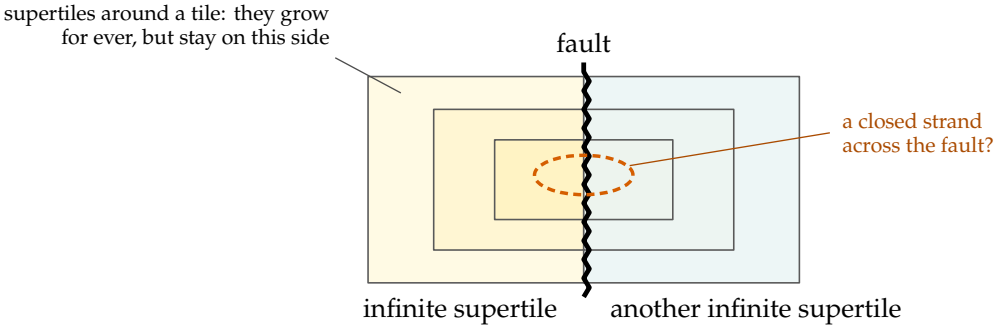


FIGURE 11. A fault, drawn schematically (supertiles as rectangles). The supertiles around a tile grow for ever but stay on one side; together they form an infinite supertile, and the boundary between two infinite supertiles, made of tile edges that no supertile crosses, is the fault. A closed strand inside one infinite supertile lies in a single supertile and is ruled out by the Signature Theorem. A closed strand crossing the fault (dashed) would lie in no single supertile; Theorem A does not cover tilings with a fault, and this paper makes no claim about them.

other infinite supertiles, built in the same way from other tiles. Two infinite supertiles do not overlap, and no supertile of any level reaches from one into another. The boundaries between them run along tile edges, and they are the fault (Fig. 11).

A closed strand that stays inside one infinite supertile lies in a single supertile, and is ruled out as before. A closed strand that crossed the fault, with pieces on both sides, would lie in no single supertile, and the argument does not reach it. So Theorem A is stated only for tilings without a fault, and this paper makes no claim about closed strands in tilings with a fault: it neither proves that there are none nor exhibits one. Theorem B is not affected, since the nest tiling has no fault (Proposition 8.4).

Tilings with faults do exist, by the usual compactness argument; we sketch it. For each $m \geq 1$, apply a proper motion to a homochiral tiling so that every tile is $\rho^k \mathrm{Tile}(1,1)+t$ ($k \in \mathbb{Z}$, $t \in \mathbb{Z}[\rho]$) and the origin is a vertex of a level-m supertile; call the result $\mathcal{T}_m$. The origin is then where three level-m supertiles meet [19, Thm. 5.1], and it lies on the boundary of a level-n supertile for every $n \leq m$. For fixed r and n, since the tiles meet edge to edge, the origin is a tile vertex, and boundedly many tiles meet the disc of radius r about the origin, there are only finitely many possibilities for these tiles, together with their grouping into level-n' supertiles for $n' \leq n$, the kinds of these supertiles and their child numbers. By a diagonal argument, some subsequence of the $\mathcal{T}_m$ has all these data eventually constant, for every r and n. The limit is a homochiral tiling whose level-n' groupings are compositions as in Proposition 2.4; by the uniqueness in Theorem 2.1(a) they are its supertiles. In it the origin lies on the boundary of a level-n supertile for every n. The finitely many tiles at the origin cover a neighbourhood of it, so they never all lie in one supertile, and they meet at least two different infinite supertiles. Tatham [20, §7.3] gives an explicit example: a tiling with two infinite supertiles side by side, with addresses $\Xi \overset{7}{\leftarrow} \Pi \overset{7}{\leftarrow} \Xi \overset{7}{\leftarrow} \cdots$ and $\Pi \overset{7}{\leftarrow} \Xi \overset{7}{\leftarrow} \Pi \overset{7}{\leftarrow} \cdots$ in his numbering of the children, almost all of which is invariant under a rotation by 60°.

(2) Are there homochiral tilings by Tile(1, 1), necessarily with faults (Theorem A), in which the decoration has a closed strand?

Other drawings and designs. The theorems concern only the pairing of ports (Definition 3.1) and the straight-through rule. They therefore apply to any drawing that realises these, including drawings on the curved Spectre, whose tilings correspond to the homochiral tilings by Tile(1, 1) [19, §2]. The method uses the decoration only through the signature of the tile (Lemma 4.4), the position of the crossover, which fixes A (Lemma 6.4), the level-1

computations (Computations 7.3 and 7.5) and, for Theorem B, the junction map (Computation 9.2). For any other decoration in which the tile has the signature at some edge, the same finite computations can be run; their outcome may differ.

Appendix A. The computations

All programs and their logs are in the ancillary files of this paper, folder `code/`. One command, `./run_all.sh`, re-runs everything the proof needs, with the quicker corroborating computations, in about twelve minutes on a four-core machine; `./run_all.sh --all` adds the corroborating computations to level 9, and takes about an hour in all (with the memory of the JavaScript programs capped at 6.5 GB). The file `code/README.md` maps every statement of this paper to the script that checks it. The Lean formalisation is in the folder `lean/`; its `build.sh` builds and checks it in about ten minutes, and its `README.md` maps the statements of this paper to the formal ones.

What the proof computes, and how. Besides the comparisons with [19, Figs. 4.1, 5.1] in the proof of Proposition 2.4, the proof uses two kinds of computation. The geometric computations on actual supertiles are the Mystic (Lemma 4.4), the level-1 data (vertex positions, ℓ_1, d_1, the position of A, and the signature of the level-1 supertiles, Computation 7.5) and the patterns of the compositions $m \to m+1$ for $m \le 3$. All of them are done in exact arithmetic by `exact/exact.py`, in Python. It represents every vertex exactly as an element $a + b\rho + c\rho^2 + d\rho^3$ of $\mathbb{Z}[\rho]$, $\rho = e^{i\pi/6}$, with integers a, b, c, d, and every motion as $z \mapsto \rho^k z + t$ (or $\rho^k \bar{z} + t$ in reflected frames). Floating point is used only to read off the sign of the area of a polygon whose area exceeds 8 in absolute value. The computations on compositions (Computations 7.3 and 9.2) use only Tables 1 to 3 and the integers ℓ_1, d_1. The program `hierarchy/compositions.py` builds each composition, follows the ribbon from every port, and checks the statement as worded. The Lean formalisation repeats Computation 7.3 and the second part of Computation 9.2 in its kernel, from the same tables, runs both negative controls as tests, and re-checks Computation 7.5 on the level-1 supertiles exported by `exact.py` (in the table below, the Lean files named are in `lean/SpectreKnot/`).

The JavaScript programs in `core/` and most of `hierarchy/` repeat the geometric computations independently and extend them: the patterns for $m \le 7$, the signature up to level 9, the merging for $b \le 6$ up to level 9. These extensions corroborate the proof but are not used in it. So do the skeleton walks, the symbolic run of the label states and its certificates (Remarks 7.8 and 9.5). They work with integers and rationals only; the integers involved are below $2 \cdot 10^7$ in absolute value, far below 2^{53}, so the parts written in JavaScript are exact as well. The inputs of the symbolic run (the pattern, the matrix M and ℓ_1) are read from the JavaScript programs and checked equal to the exact ones. The programs share no code with `exact.py`, although both implement the key-point construction of [19, App. A] and so the same table of placements. They work in double-precision floating point and identify points by rounding coordinates to 10^{-2} or 10^{-3}. We do not bound their rounding error. A separate check (`crosscheck/overlap.js`), which builds the supertiles of levels 1 to 7 in the same way, finds distinct vertices at distance at least 0.89, and points that it identifies within 4×10^{-12} of each other; and up to level 9 every rim found has exactly E_m edges (Proposition 6.2) and every rim vertex has degree 2, which a wrongly identified point would break. All programs compose the ribbon pairings with Lemma 4.2 and nothing else. (In the programs the sign ω_m is called `beta`.)

Sizes. The script `hierarchy/sizes.py` computes these numbers from Table 1 and Proposition 6.2. They agree with the supertiles built by the programs (tile counts up to level 7, rim lengths up to level 9).

level m	1	2	3	4	5	6	7	8	9
tiles in a Δ	9	71	559	4401	34649	272791	2147679	16908641	133121449
rim $E_m(\Delta)$	46	182	758	3198	13534	57318	242790	1028462	4356622
shortest side β_m	5	15	57	235	989	4183	17713	75027	317813

What was computed.

statement	computation	script
Definition 3.1 and Lemma 4.4	the tile's pairing; the Mystic	`core/tile.js`, `exact/exact.py`
Proposition 2.4, Remark 2.5	vertices of levels 1–6 (points where three supertiles meet); the pattern of compositions $m \to m+1$, $m = 1, \ldots, 7$, all identical; exact for $m \leq 3$	`hierarchy/vertices.js`, `combin.js`, `exact/exact.py`
Proposition 2.4 (base of $(*)$)	the level-1 supertiles as hat/turtle clusters, fitted to [19, Fig. 4.1] (dots, outline, shading; labels read by hand)	`crosscheck/fig41_compare.py`, `fig41_render.py`
Table 1, Proposition 2.4 (base of $(*)$)	Table 1 and every tile of every supertile, levels 0–4, against Kaplan's implementation [14] (downloaded when run, not included); from its tiles alone, the level-1 vertices, ℓ_1 and d_1; two altered copies rejected	`crosscheck/kaplan_compare.py`, `kaplan_run.js`
Remark 2.5	agreement with an encoding of [19, Fig. 5.1], all nine kinds; its dots re-found from the PDF; two corrupted encodings rejected	`crosscheck/fig51_compare.py`, `fig51_raster.py`
Remark 2.5	tiles of every supertile, levels 1–7, have disjoint interiors and fill its rim	`crosscheck/overlap.js`
Propositions 6.1 and 6.2, Corollary 6.3, and Lemma 6.4	letters, recurrence, lengths at levels 1–8, closed forms	`hierarchy/recur.js`, `letters.py`
Computation 7.3	the 18 compositions $C_T(\ell_1, d_1, \pm 1)$ from the tables, every port followed; negative control	`hierarchy/compositions.py`; Lean: `CompFull.lean`
Computation 7.5	signature of the level-1 supertiles: exact, and in floating point (both also at levels 2–3, and 4)	`exact/exact.py`, `hierarchy/checks.js`; Lean: `PatchSig.lean`
Computation 9.2	the junction map in $C_\Delta(\ell_1, d_1, \pm 1)$; negative control	`hierarchy/compositions.py`; Lean: `StepMap.lean`
Remark 7.8 (1), Remark 9.5 (2)	the compositions and the junction map at $(\ell_m, d_m, \pm 1)$, $m \leq 4$; every single move from (ℓ_1, d_1)	`hierarchy/compositions.py`
Remark 7.8 (1)	the moves of Lemma 7.2 followed back from levels 2 and 3, checked step by step	`crosscheck/chain_check.py`
Remark 9.5 (2)	Table 6 on the actual blocks $N_1, \ldots, N_4$, exact arithmetic; negative control	`crosscheck/junction_actual.py`
Remark 7.8 (3)	the 18 skeleton walks; parity lemma for skeletons; equal to the actual skeletons at levels 2–9 (the computed skeletons record slightly more than Definition 5.3, e.g. rim-run parities)	`hierarchy/induction.js`
Proposition 5.4 (check)	reduced walk equals direct composition at all 1,424 parent ports, levels 1–8	`hierarchy/checks.js`
Remark 9.5 (3)	symbolic label computation, 1,068 + 1,038 obligations; certificates	`hierarchy/certnest.js`, `certify_fib.py`, `certify.py`
Table 5, Remark 9.5 (1)	strand following in exact arithmetic for $b \leq 3$ to level 7; in floating point for $b \leq 6$ to level 9; label computation for $b = 1, 2, 3$	`exact/exact.py`, `core/hier2.js`, `core/nest.js`
corroboration	signature and skeleton periodicity at levels 1–9, all kinds; 0 closed strands at every level	`core/reduce.js`, `hierarchy/synth.js`
Section 3 (drawing)	the drawn ribbons realise the pairing, levels 1–3; shared edges glue e_i to e_j with $i + j$ odd, levels 1–4	`drawing/checkmodel.js`, `exact/exact.py`
Remark 3.6	the drawn pieces: containment, the single crossing, angles at the edges	`drawing/drawcheck.py`

References

1. Michael Baake and Uwe Grimm, *Aperiodic order. Volume 1: A mathematical invitation*, Encyclopedia of Mathematics and its Applications, vol. 149, Cambridge University Press, Cambridge, 2013.
2. George Bain, *Celtic art: The methods of construction*, William MacLellan, Glasgow, 1951.
3. bohemian-miser, *Spectre: playing with the spectre tiling*, https://github.com/bohemian-miser/Spectre, file `docs/FASS_THEOREM.md` (accessed 8 October 2026), 2026.

4. Joseph I. Cline, *Interlace patterns emerging in a Penrose-type Islamic design*, Proceedings of Bridges 2024: Mathematics, Art, Music, Architecture, Culture, 2024, pp. 163–170.
5. Peter R. Cromwell, *Celtic knotwork: mathematical art*, The Mathematical Intelligencer **15** (1993), no. 1, 36–47.
6. _____, *Knots and links*, Cambridge University Press, Cambridge, 2004.
7. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
8. Gwen Fisher and Blake Mellor, *On the topology of Celtic knot designs*, Bridges: Mathematical Connections in Art, Music, and Science (Winfield, Kansas), 2004, pp. 37–44.
9. Felix Flicker, Steven H. Simon, and S. A. Parameswaran, *Classical dimers on Penrose tilings*, Physical Review X **10** (2020), 011005.
10. Chaim Goodman-Strauss, *Aperiodic hierarchical tilings*, Foams and Emulsions (J. F. Sadoc and N. Rivier, eds.), Springer Netherlands, Dordrecht, 1999, pp. 481–496.
11. Branko Grünbaum and G. C. Shephard, *Tilings and patterns*, second ed., Dover, 2016.
12. W. Patrick Hooper, *Truchet tilings and renormalization*, arXiv:1103.6046, 2011.
13. Slavik Jablan, Ljiljana Radović, Radmila Sazdanović, and Ana Zeković, *Knots in art*, Symmetry **4** (2012), no. 2, 302–328.
14. Craig S. Kaplan, *The spectre: a chiral aperiodic monotile*, https://cs.uwaterloo.ca/~csk/spectre/.
15. Stefan Pautze, *Space-filling, self-similar curves of regular pentagons, heptagons and other n-gons*, Proceedings of Bridges 2021: Mathematics, Art, Music, Architecture, Culture, 2021, pp. 157–164.
16. Gábor Pete, *Corner percolation on $\mathbb{Z}^2$ and the square root of 17*, Annals of Probability **36** (2008), no. 5, 1711–1747.
17. Shobhna Singh, Jerome Lloyd, and Felix Flicker, *Hamiltonian cycles on Ammann–Beenker tilings*, Physical Review X **14** (2024), 031005.
18. David Smith, Joseph Samuel Myers, Craig S. Kaplan, and Chaim Goodman-Strauss, *An aperiodic monotile*, Combinatorial Theory **4** (2024), no. 1, Paper No. 6, arXiv:2303.10798.
19. _____, *A chiral aperiodic monotile*, Combinatorial Theory **4** (2024), no. 2, Paper No. 13, arXiv:2305.17743.
20. Simon Tatham, *Finite-state transducers for substitution tilings*, arXiv:2512.16595, 2025.

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